



ROSS UNIVERSITY
SCHOOL OF MEDICINE

PELVIS AND PERINEUM 2

Part 2

Male External Genitalia and Pelvic Organs

Recommended Reading:

Grays Anatomy for Students, 4th Edition: Pages 453 - 469, 499 - 505.

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LEARNING OBJECTIVES

By the end of this lecture and after the recommended reading you should be able to:

- Understand the male external genitalia and compare it to the female external genitalia.
- Understand the positioning and relationships of pelvic organs in the male.
- Be able to identify different parts of pelvic organs in the male pelvis.
- Relate clinical concepts to basic science knowledge as learned in the lecture.

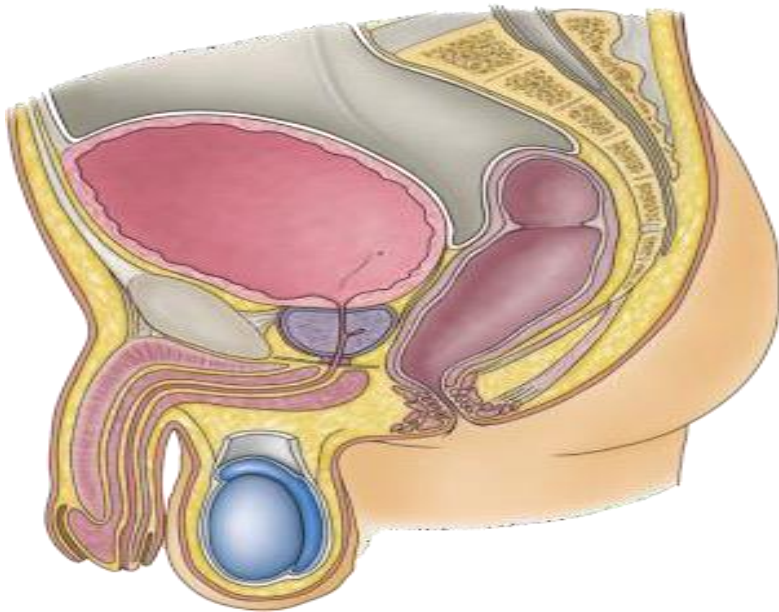
RECAP: TIPS FOR LEARNING PELVIC ANATOMY

- When viewing a sagittal section of a pelvis, look for the pubic symphysis and sacrum to get oriented to anterior vs. posterior aspects.
- Look at the genital region to determine sex/gender. If you see a uterus then the specimen is female. BUT... if a specimen appears to have no uterus, do not assume it must be a *male*... always use a second type of verification method by observing what external genitalia are present.
- Important to note the location of organs and structures within the pelvis and their relative positioning to one another. (Surgeons have to be aware of this).
- Learn the anatomy as being relevant to your physical exam procedures, diagnostic procedures, and treatments for your future patients.

GENITALIA AND ORGANS OF THE PELVIS

URINARY ORGANS

- Ureter
- Urinary Bladder
- Urethra



Male

GENITAL ORGANS

Males:

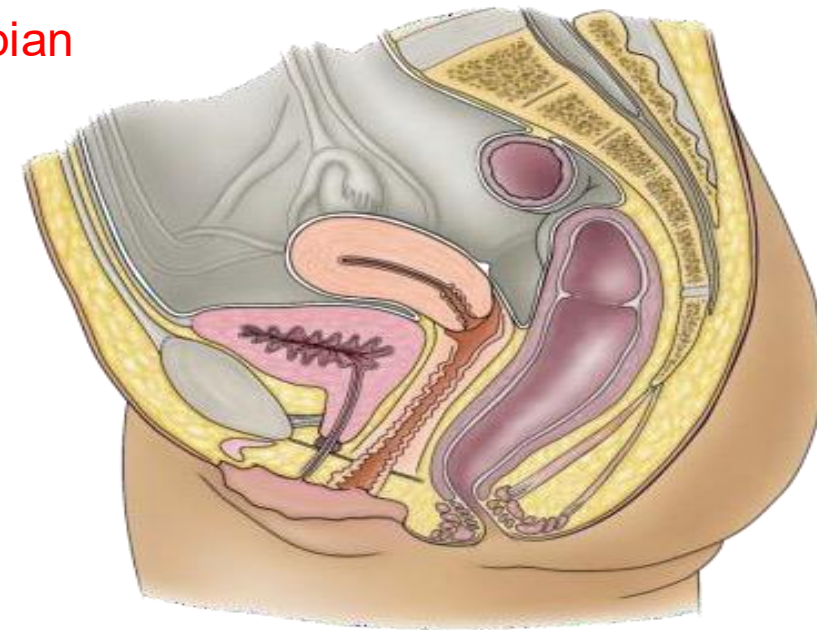
- Vas Deferens
- Testis
- Epididymis
- Seminal Vesicles
- Prostate Gland
- Penis

Females:

- Vulva
- Clitoris
- Vagina
- Uterine/Fallopian Tube
- Uterus
- Cervix
- Ovary

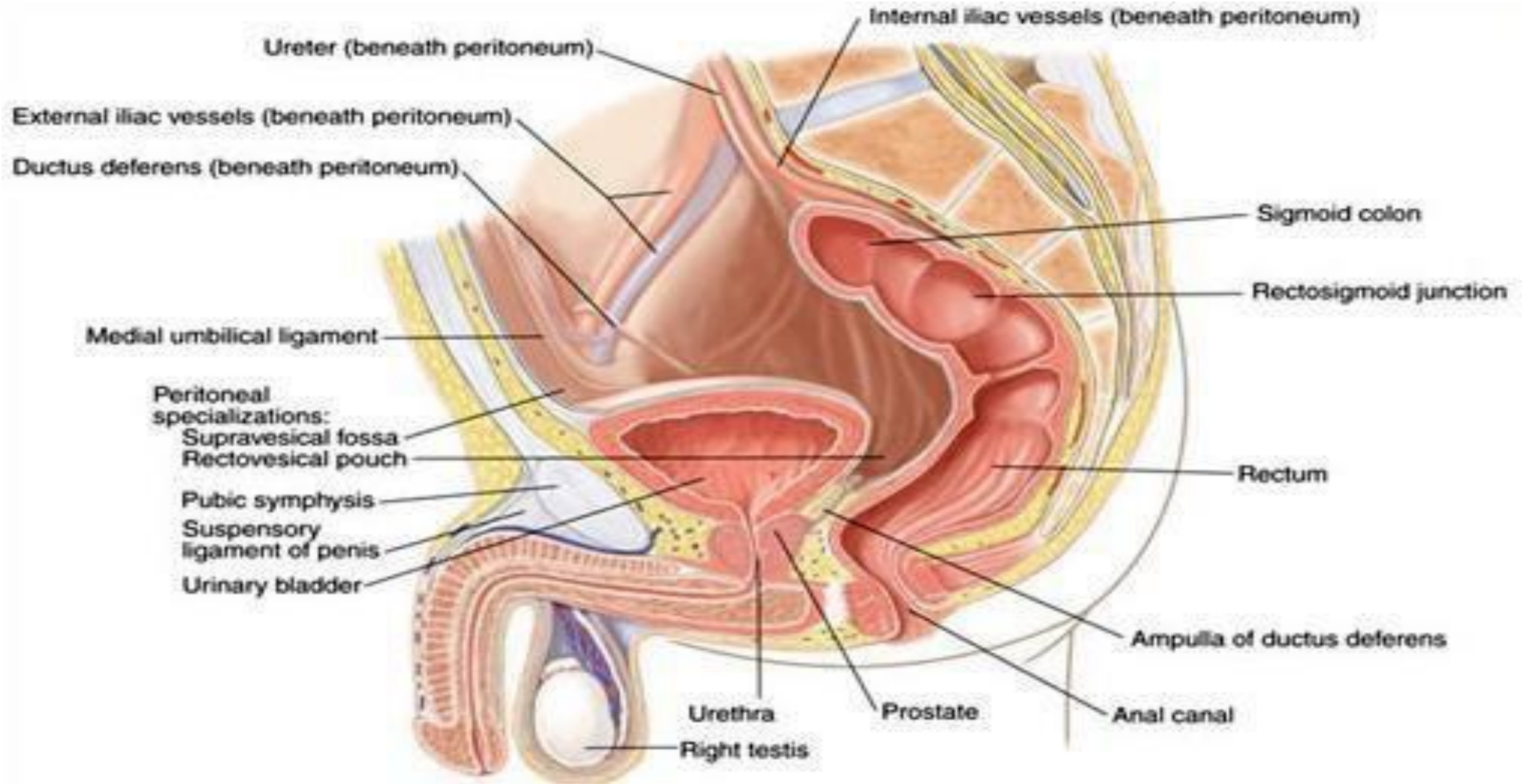
DIGESTIVE ORGANS

- Parts of the colon
- Rectum
- Anal Canal



Female

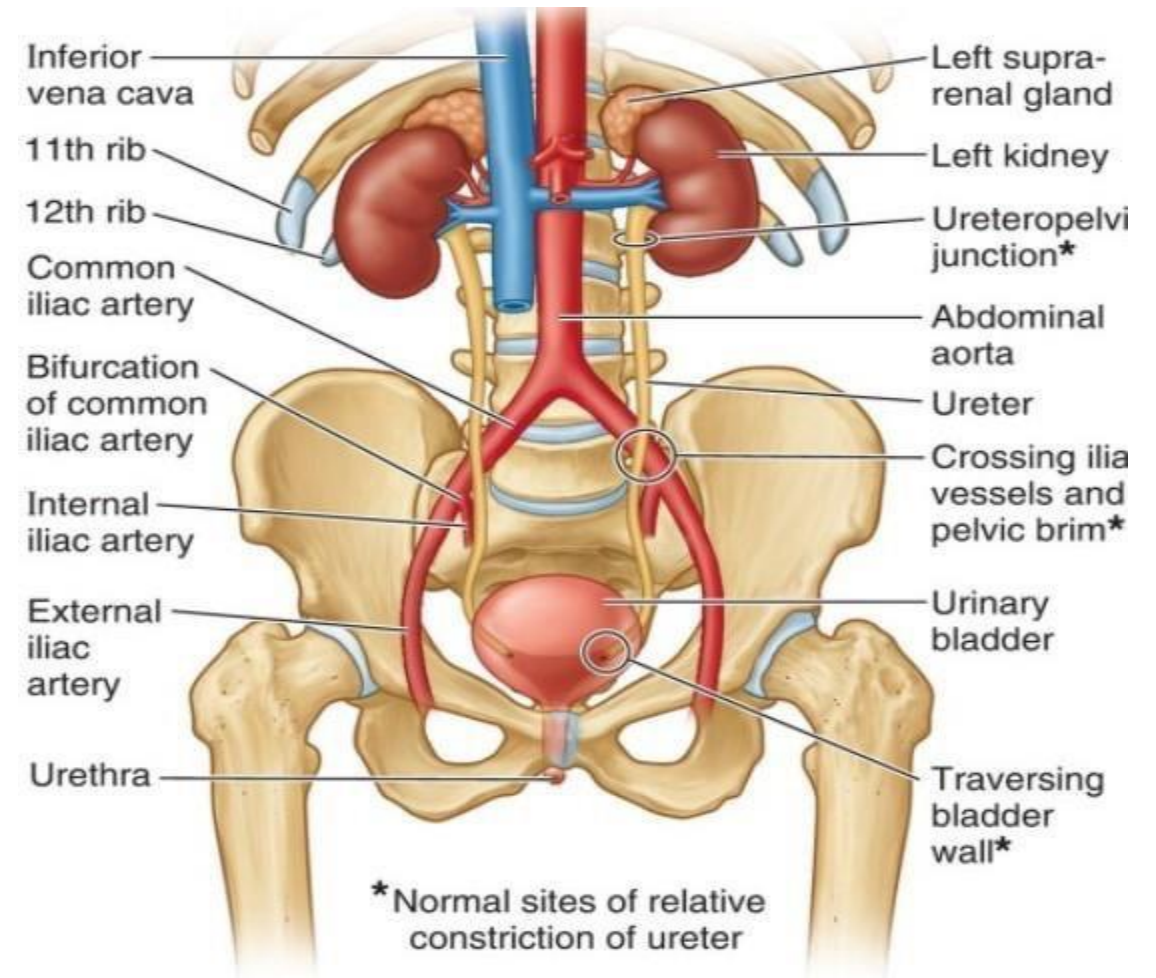
STRUCTURES OF THE MALE PELVIS



*Male
pelvis,
sagittal
section*

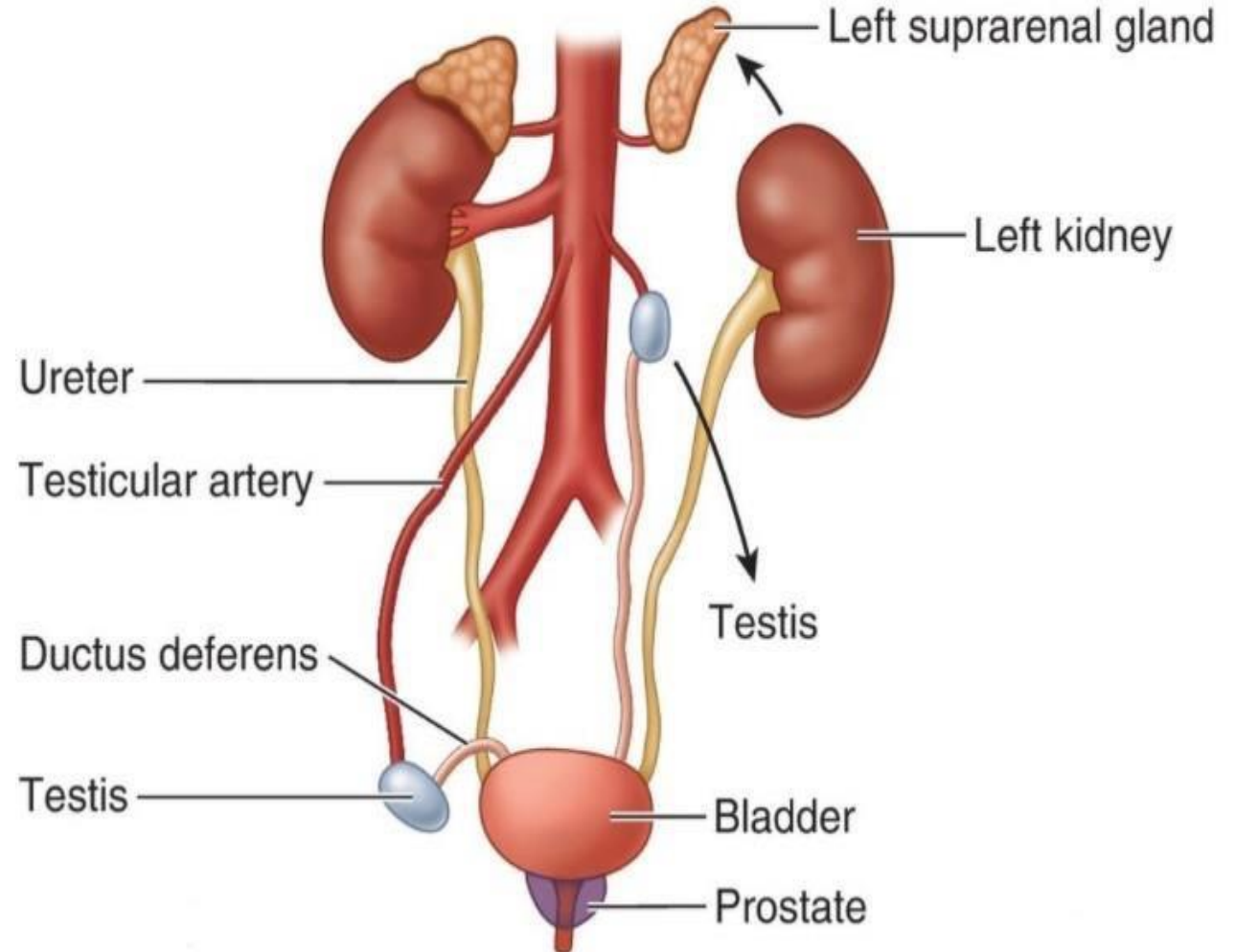
URINARY ORGANS

- The pelvic urinary organs include:
 - Pelvic portions of the ureters
 - Urinary bladder
 - Urethra



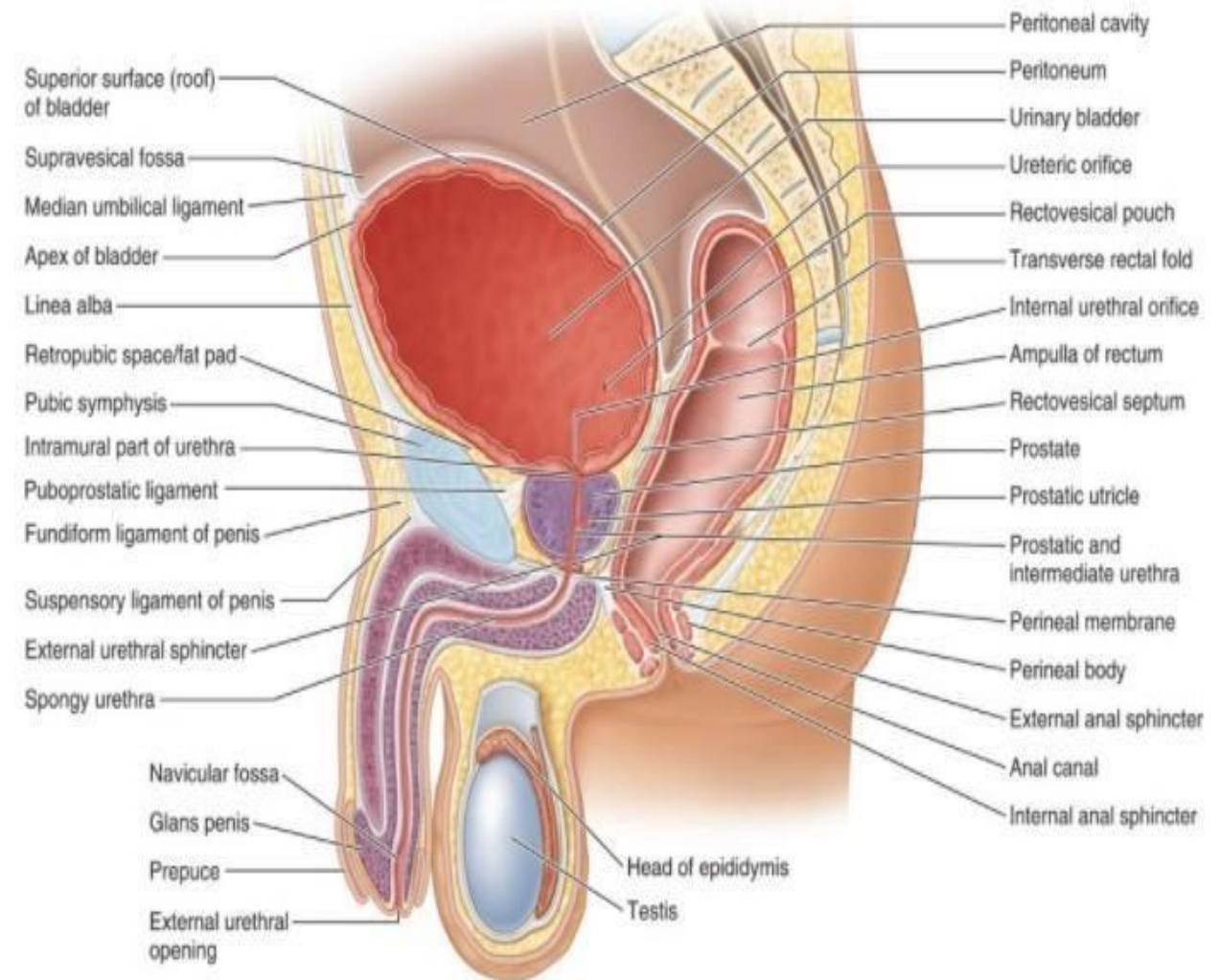
URETERS AND RELATION TO THE DUCTUS DEFERENS

- **Ureters:** 2 thick, muscular tubes which act to transport urine from the kidney to the bladder.
- In males, the only structure that passes between the ureter and the peritoneum is the ductus deferens.
- The ductus deferens crosses the ureter within the ureteric fold of peritoneum.
- The ureter lies posterolateral to the ductus deferens and enters the posterosuperior angle of the bladder, superior to the seminal gland.



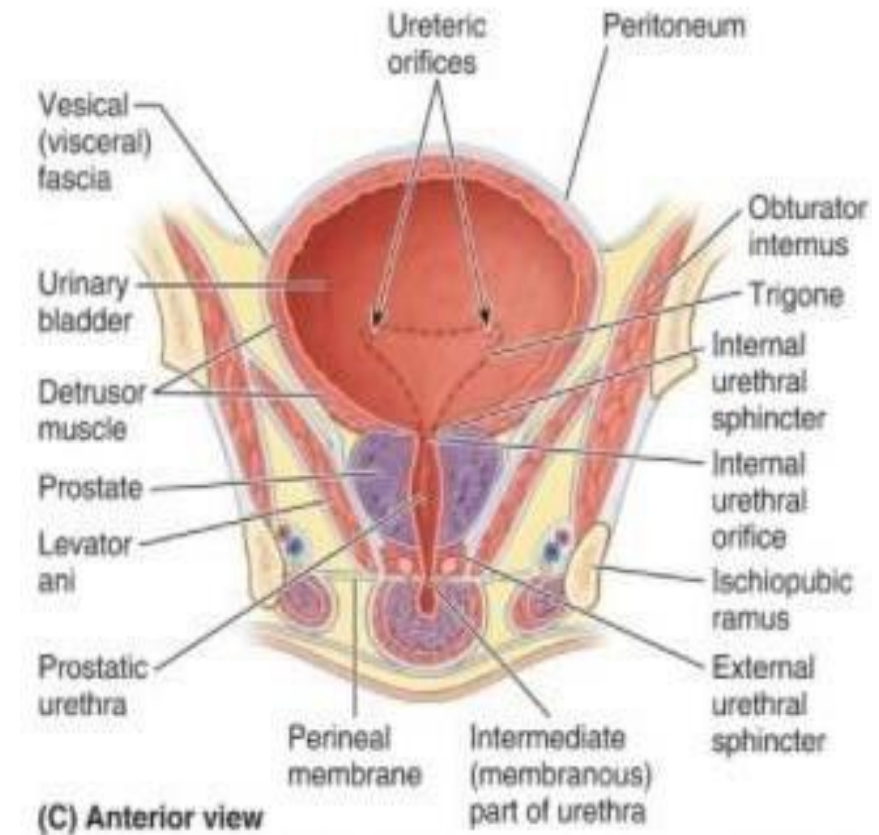
THE URINARY BLADDER

- See slide in Female External Genitalia and Pelvic Organs Lecture!



THE URINARY BLADDER

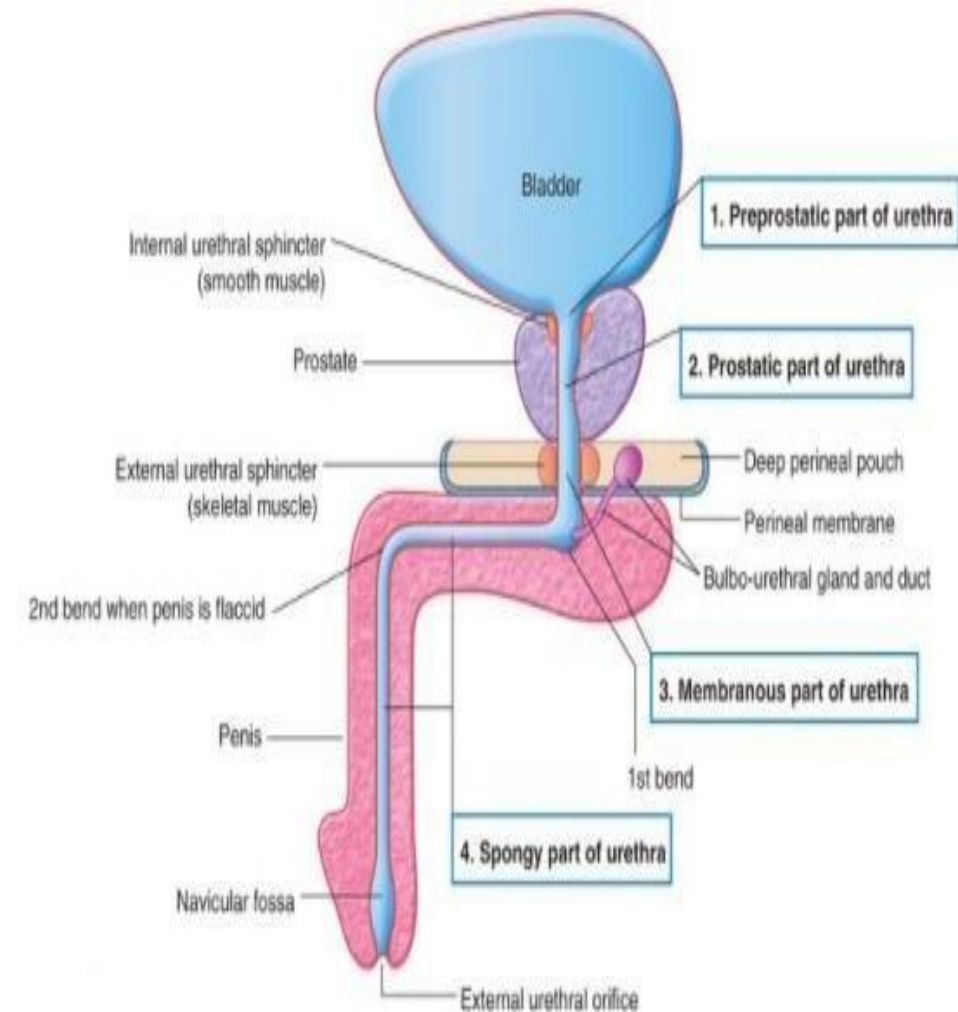
- Musculature of the bladder plays a key role in storage and emptying of urine.
 - **Detrusor Muscles**
 - ▶ Specialized smooth muscle fibers which makes up the wall of the bladder.
 - ▶ Contraction of the muscles causes micturition.
 - **Internal Urethral Sphincter**
 - ▶ Involuntary muscle which surrounds the neck of the bladder and the fibromuscular tissue of the prostate.
 - ▶ Contracts during ejaculation to prevent retrograde ejaculation (ejaculatory reflux).
 - **External Urethral Sphincter**
 - ▶ Skeletal muscle
 - ▶ Under voluntary control (innervated by the pudendal nerve).



The two ureters open into the bladder from the posterior aspect.

THE URETHRA

- Muscular tube (18-22cm long in males) that transports urine and semen.
- The urethra is divided into 4 parts:
 - **Preprostatic (intramural) part of the urethra**
 - ▶ Extends from the base of the bladder to the prostate.
 - ▶ Associated with the internal urethral sphincter.
 - **Prostatic urethra**
 - ▶ Surrounded by the prostate.
 - ▶ Receives the ejaculatory ducts and the prostatic ducts.
 - ▶ It is the widest and most dilatable portion of the urethra
 - **Membranous urethra**
 - ▶ Passes through the deep perineal pouch.
 - ▶ Surrounded by skeletal muscle of the external urethral orifice.
 - ▶ It is the narrowest and least dilatable portion of the urethra
 - **Penile (bulbous; spongy) urethra**
 - ▶ Passes through the bulb and corpus spongiosum of the penis, ending at the external urethral orifice.
 - ▶ Receives the bulbourethral glands proximally.
 - ▶ In the glans (head) of the penis, the urethra dilates to form the navicular fossa.

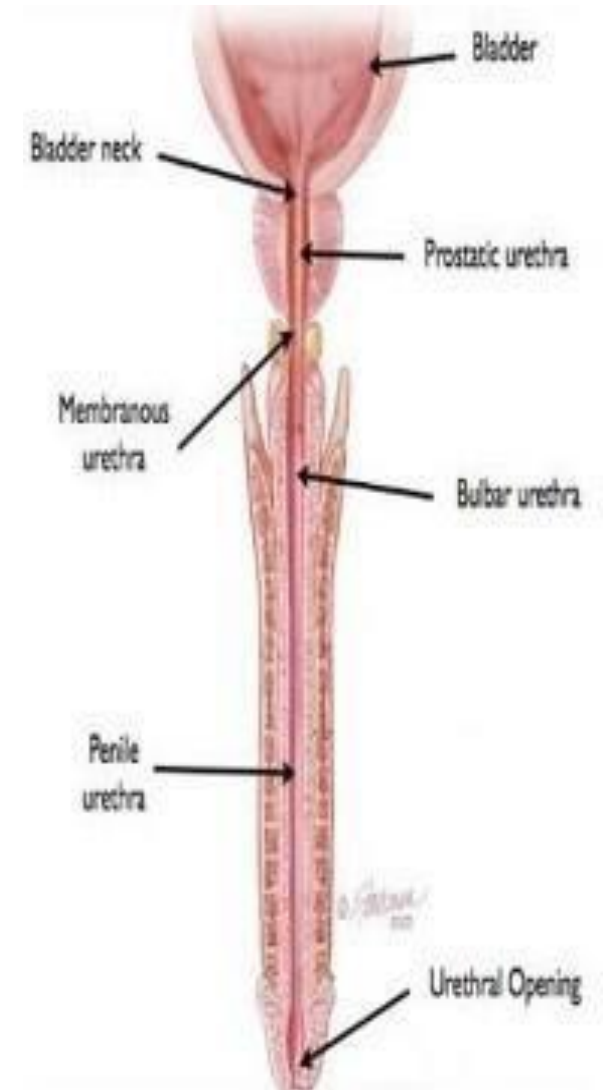


THE URETHRA

TABLE 6.6. PARTS OF MALE URETHRA

Part	Length ^a	Location/Disposition	Features
Intramural (preprostatic) part	0.5–1.5 cm	Extends almost vertically through the neck of the bladder	Surrounded by internal urethral sphincter; diameter and length vary, depending on whether bladder is filling or emptying
Prostatic urethra	3.0–4.0 cm	Descends through anterior prostate, forming a gentle, anteriorly concave curve; is bounded anteriorly by a vertical trough-like part (rhabdosphincter) of external urethral sphincter	Widest and most dilatable part; features urethral crest with seminal colliculus, flanked by prostatic sinuses into which prostatic ducts open; ejaculatory ducts open onto colliculus; hence, urinary and reproductive tracts merge in this part
Intermediate (membranous) part	1.0–1.5 cm	Passes through deep perineal pouch, surrounded by circular fibers of external urethral sphincter; penetrates perineal membrane	Narrowest and least distensible part (except for external urethral orifice)
Spongy urethra	~15 cm	Courses through corpus spongiosum; initial widening occurs in the bulb of the penis; widens again distally as navicular fossa (in glans penis)	Longest and most mobile part; bulbo-urethral glands open into bulbous part; distally, urethral glands open into small urethral lacunae entering the lumen of this part

^aLengths are provided for purposes of comparison—students should not memorize these lengths.

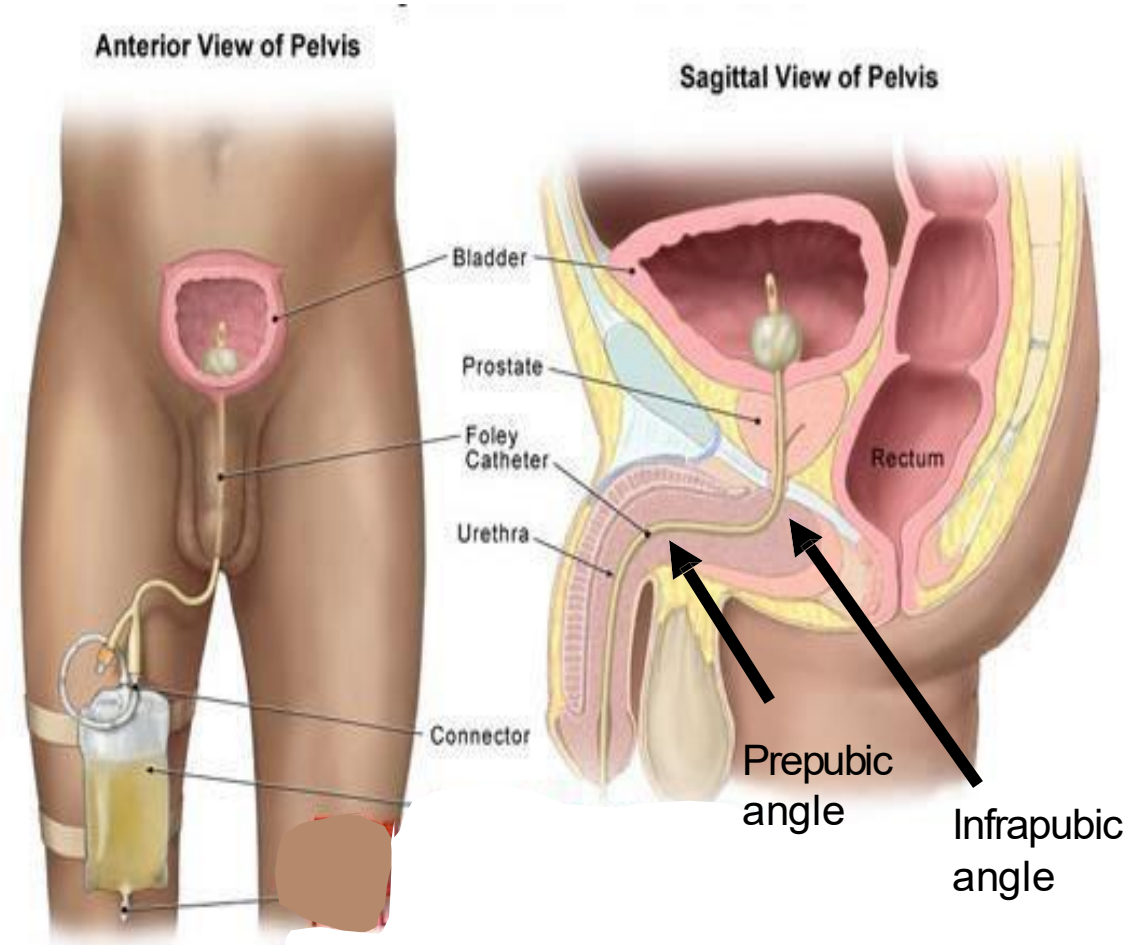


NEUROVASCULAR SUPPLY AND LYMPHATIC DRAINAGE OF THE URETHRA

- **Arterial Supply:**
 - **Prostatic urethra:** Inferior vesical artery (branch of the internal iliac artery that supplies the lower part of the bladder).
 - **Membranous urethra:** Bulbourethral artery (branch of the internal pudendal artery).
 - **Penile urethra:** Supplied directly by the dorsal artery of the penis (a branch of the internal pudendal artery).
- **Venous supply:**
 - Veins from the proximal two parts drain into the prostatic venous plexus.
 - Veins from the distal parts of drain into the veins accompanying the arteries.
- **Nerve supply:**
 - **Prostatic plexus** (contains a mixture of parasympathetic, sympathetic and visceral afferent fibers).
- **Lymphatic Drainage:**
 - **Prostatic and membranous portion of the urethra:** obturator and internal iliac nodes.
 - **Penile urethra:** Deep and superficial inguinal nodes

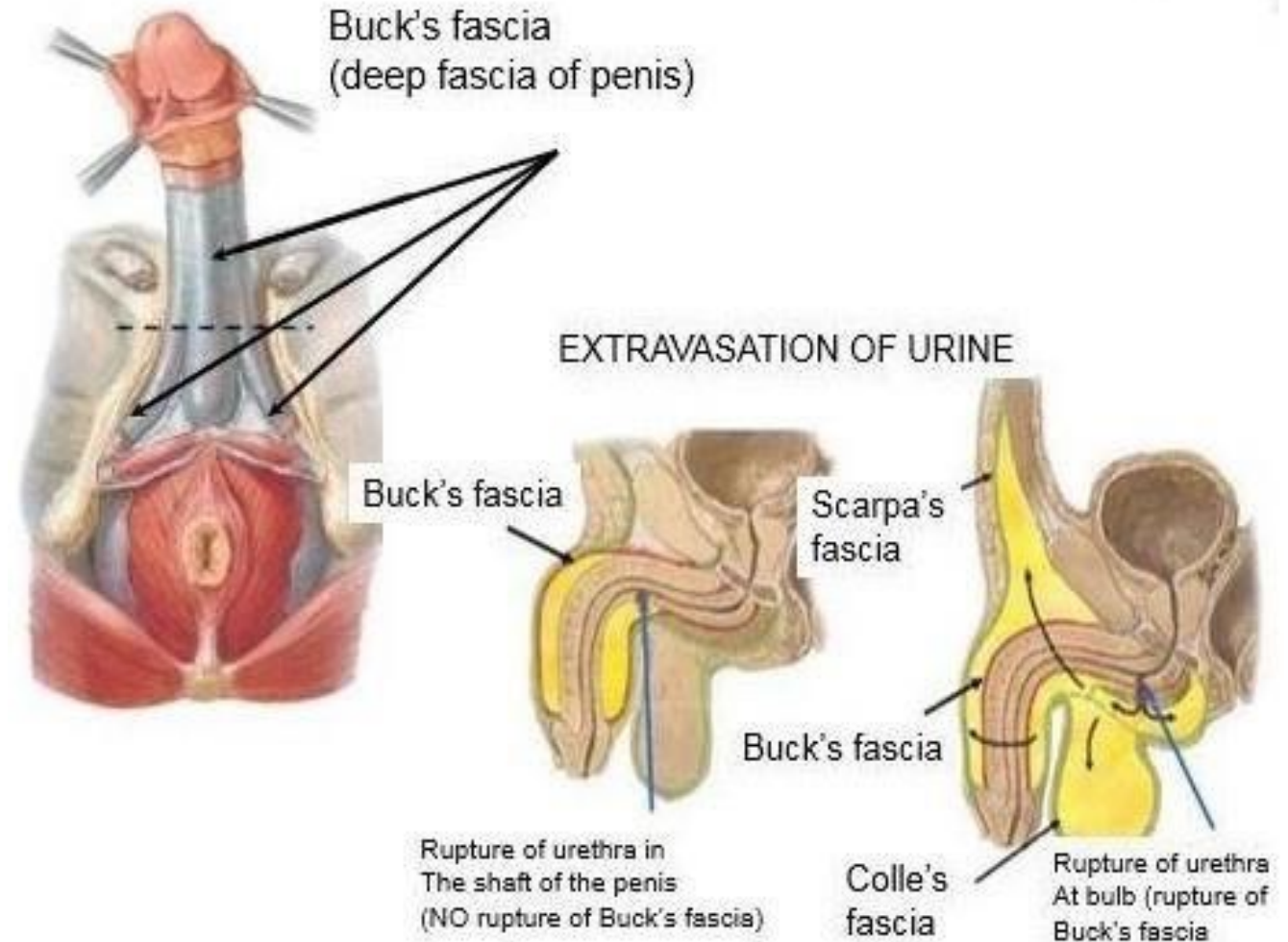
CLINICAL CORRELATE: MALE CATHETERIZATION

- Urinary catheterization is the process of inserting a tube through the urethra into the bladder to drain urine.
- Indications:
 - Monitor urine output.
 - Urinary retention.
 - Obtain an uncontaminated sample of urine.
 - Irrigate the bladder.
- In males, catheterization is more complex because there are two angles to consider:
 - Infrapubic angle
 - Prepubic angle: Can be diminished by holding the penis upwards during urinary catheterization.
- There are three constrictions in the male urethra:
 - Internal urethral sphincter
 - External urethral sphincter
 - External urethral orifice

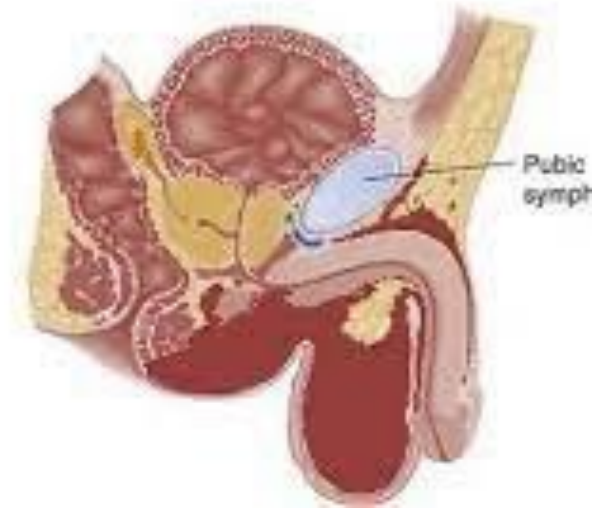
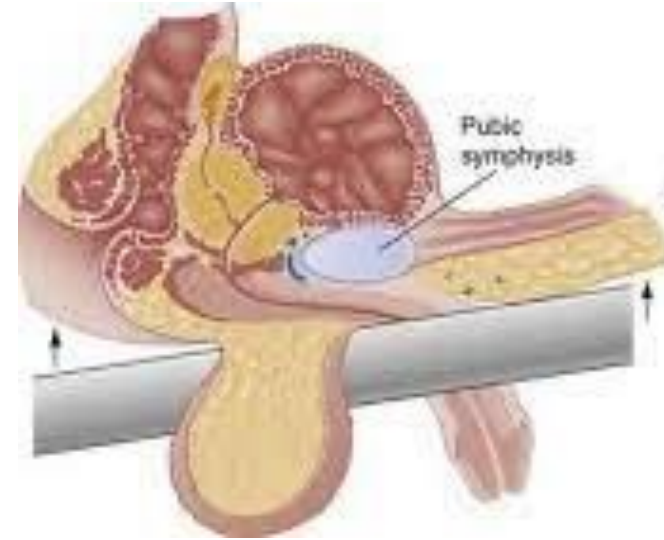
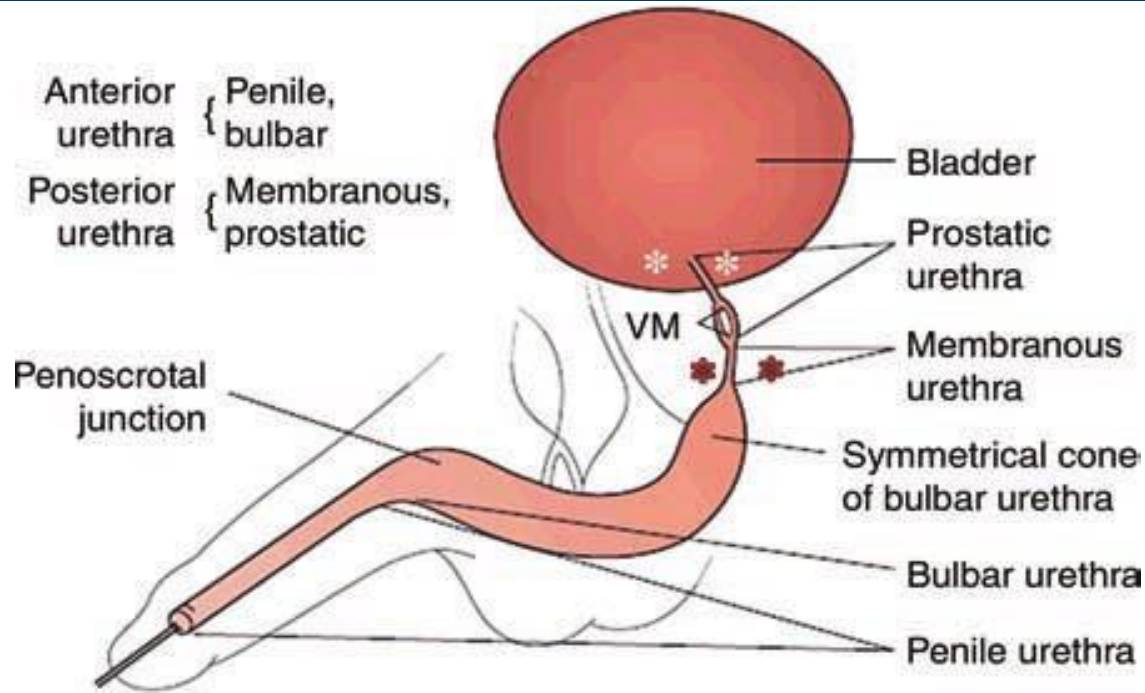


CLINICAL CORRELATE: EXTRAVASATION OF URINE

- An interruption of the urethra leads to a collection of urine in other cavities such as the scrotum of penis in males.
- If the injury to the urethra leaves Buck's fascia intact, urine will collect in the penis deep to Buck's fascia.
- If the injury to the urethra is accompanied by a tear of Buck's fascia then extravasated urine and blood can accumulate in the superficial perineal space, spread inferiorly to the scrotum, anteriorly around the penis (outer to Buck's fascia) and superiorly into the anterior abdominal wall.



STRADDLE INJURY



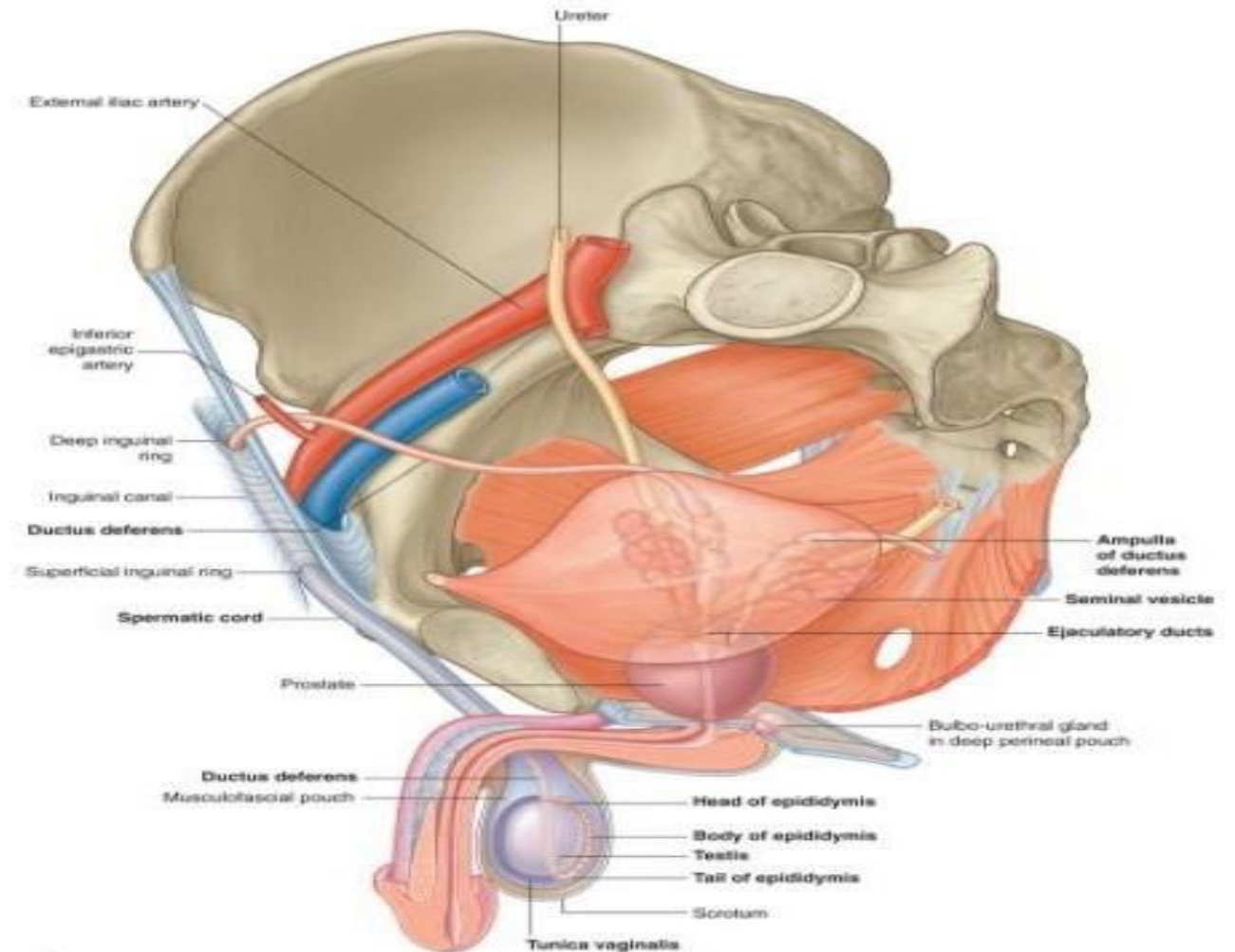
Anterior portion of urethra (external/outside part) gets ruptured

Deeper **posterior** portion may be injured in other instances e.g Motor vehicle accidents, open-book fractures. The membranous portion gets injured in these instances. Urine won't collect in the superficial perineal space, but it will collect in deep perineal space.



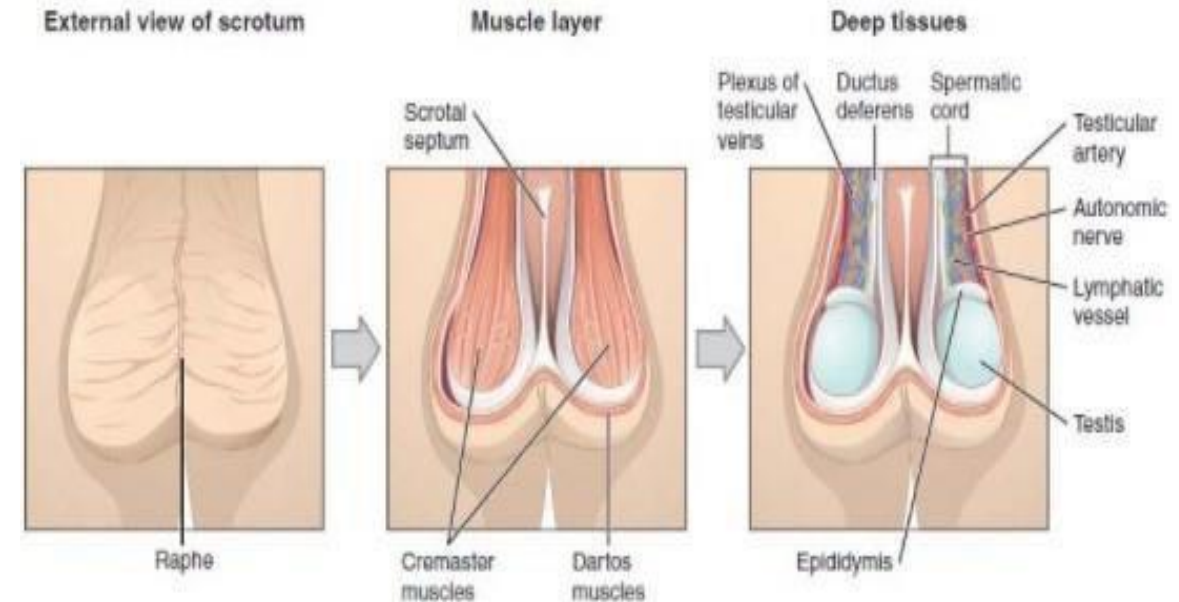
MALE INTERNAL GENITALIA IN THE PELVIS

- The male internal genitalia include:
 - The Scrotum
 - The Testes
 - The Epididymis
 - The Ductus Deferens
 - The Seminal Vesicles
 - The Prostate Gland
 - The Bulbourethral Glands
 - The Penis



THE SCROTUM

- A dual-chambered, fibromuscular cutaneous sac located between the penis and the anus
- Contains three major paired organs:
 - Testis: Site of sperm production
 - Epididymis: Stores sperm
 - Spermatic cord: Collection of muscle fibers, vessels, nerves and ducts that run to and from the testes.
- Situated postero-inferior to the penis and inferior to the pubic symphysis.
- Within the scrotum, the dartos muscle is a sheet of smooth muscle situated directly underneath the skin.
 - Regulates the temperature of the scrotum by wrinkling the skin to decrease the surface area and reduce heat loss.

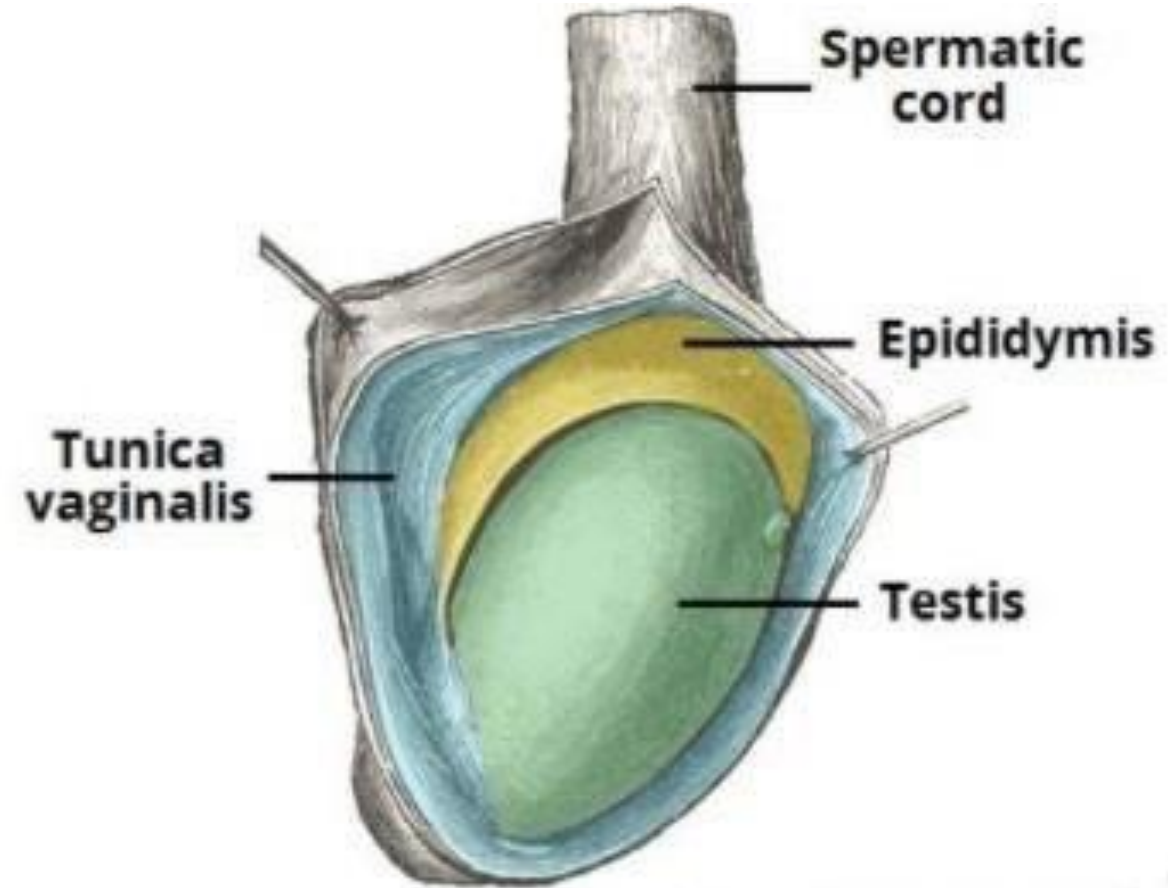


NEUROVASCULAR SUPPLY AND LYMPHATIC DRAINAGE OF THE SCROTUM

- **Arterial Supply:**
 - Anterior scrotal artery, a branch of the external pudendal artery: supplies the anterior aspect of the scrotum.
 - Posterior scrotal artery, a branch of the internal pudendal artery: supplies the posterior aspect of the scrotum.
- **Venous Drainage:**
 - Scrotal veins which drain into the external pudendal veins.
- **Lymphatic Drainage:**
 - Superficial inguinal lymph nodes.
- **Nerves**
 - Cutaneous innervation to the scrotum is supplied by several nerves, according to topography:
 - ▶ Anterior and anterolateral aspect of the scrotum is innervated by anterior scrotal nerves which are derived from the genital branch of the genitofemoral nerve and the ilioinguinal nerves.
 - ▶ Posterior aspect of the scrotum is innervated by posterior scrotal nerves, derived from perineal branches of the pudendal nerve and the posterior femoral cutaneous nerve.

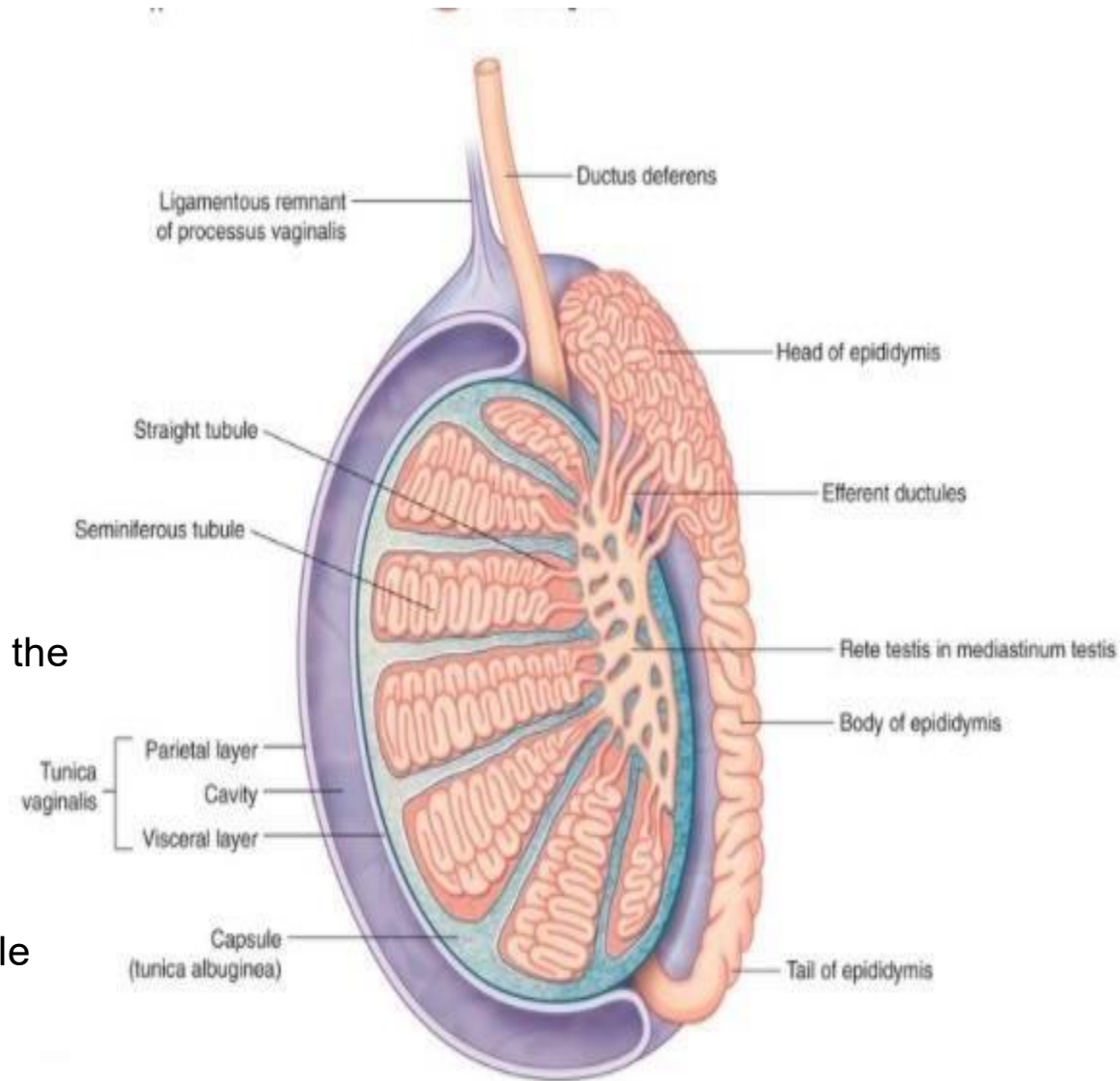
THE TESTES AND EPIDIDYMISS

- The testes and epididymis are paired structures located within the scrotum. The epididymis is located on the posterolateral aspect of each testicle.
- The testes are the site of sperm production and hormone synthesis.
- The testicles are suspended from the abdomen by the spermatic cord and the left testicle usually lies lower than the right testicle.
- The testes are originally located in the posterior abdominal wall and during embryonic development they descend down the abdomen and through the inguinal canal to reach the scrotum.



THE TESTES AND EPIDIDYMIS

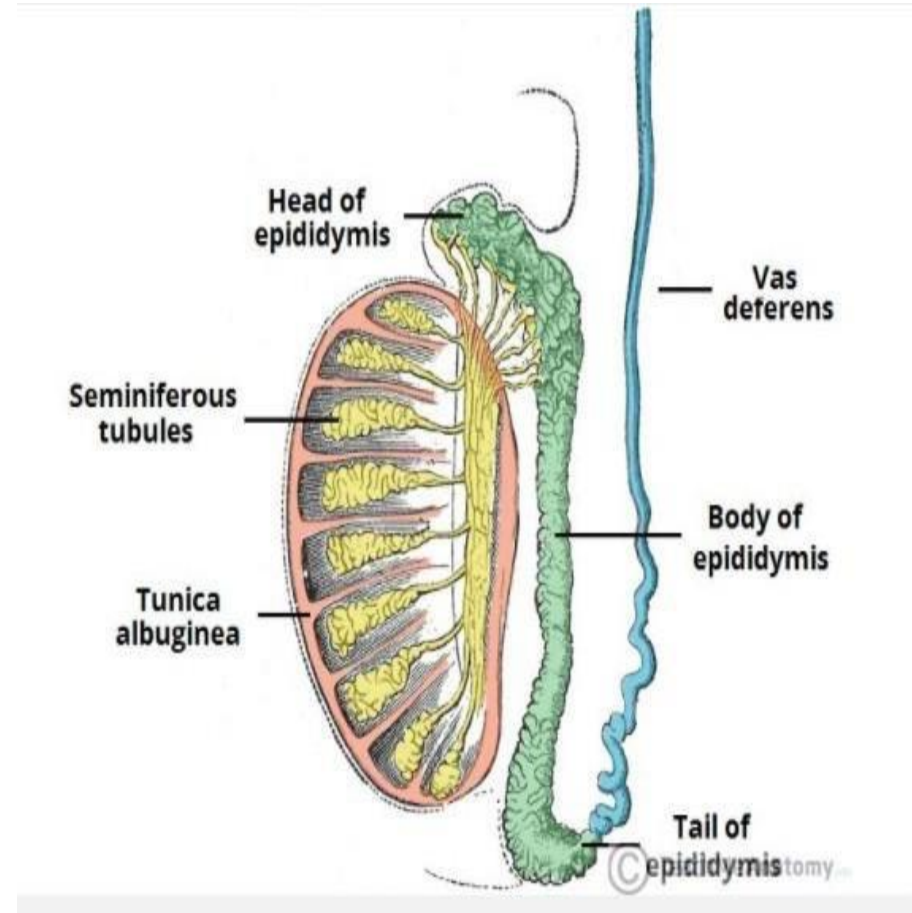
- Testes are ovoid in shape and consist of a series of lobules each containing seminiferous tubules supported by interstitial tissue.
 - Seminiferous tubules produce spermatozoa and contain Sertoli cells.
 - Leydig cells are found in the interstitial tissue and produce testosterone.
- Sperm produced in the seminiferous tubules travel through the tubules and collect in the rete testes. Efferent tubules then transport the sperm from the rete testes to the epididymis for storage and maturation.
- Within the scrotum the testes are covered by the tunica vaginalis
 - Tunica vaginalis is a closed peritoneal sac partially surrounding the testes which consists of the **visceral and parietal layer of the tunica vaginalis**.
- The tunica albuginea is a fibrous capsule that encloses the testes to protect the testicular parenchyma.
 - The tunica albuginea penetrates the parenchyma of each testicle dividing it into lobules.



THE TESTES AND EPIDIDYMIS

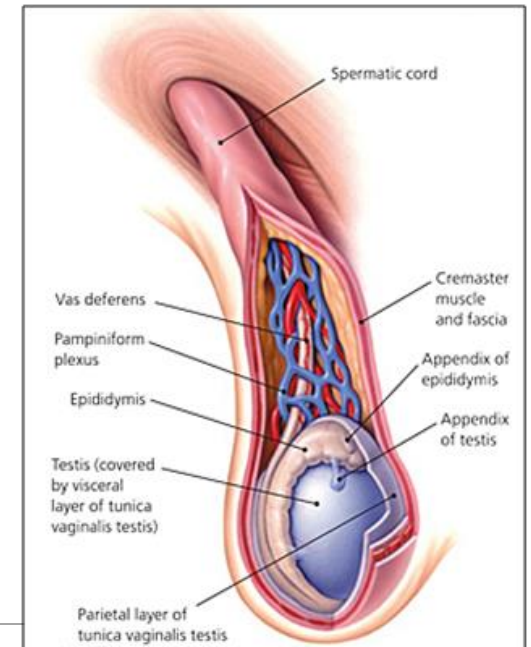
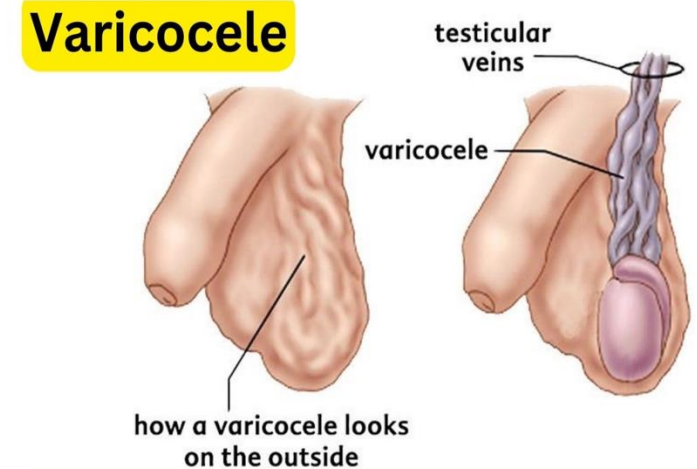
- The epididymis is an elongated structure on the posterior surface of the testis which consists of a single heavily coiled duct and is divided into three parts:
 - **Head**
 - ▶ Most proximal part of the epididymis.
 - ▶ Composed of lobules formed by the coiled ends of 12-14 efferent ductules.
 - **Body**
 - ▶ Formed by heavily coiled ducts of the epididymis.
 - **Tail**
 - ▶ The most distal part of the epididymis
 - ▶ Continuous with the vas deferens, which transports sperm to the prostatic portion of the urethra for ejaculation.

*During passage through the epididymis the spermatozoa acquire the ability to move and fertilize an oocyte.



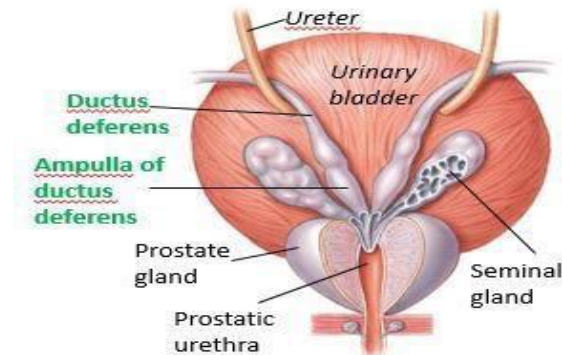
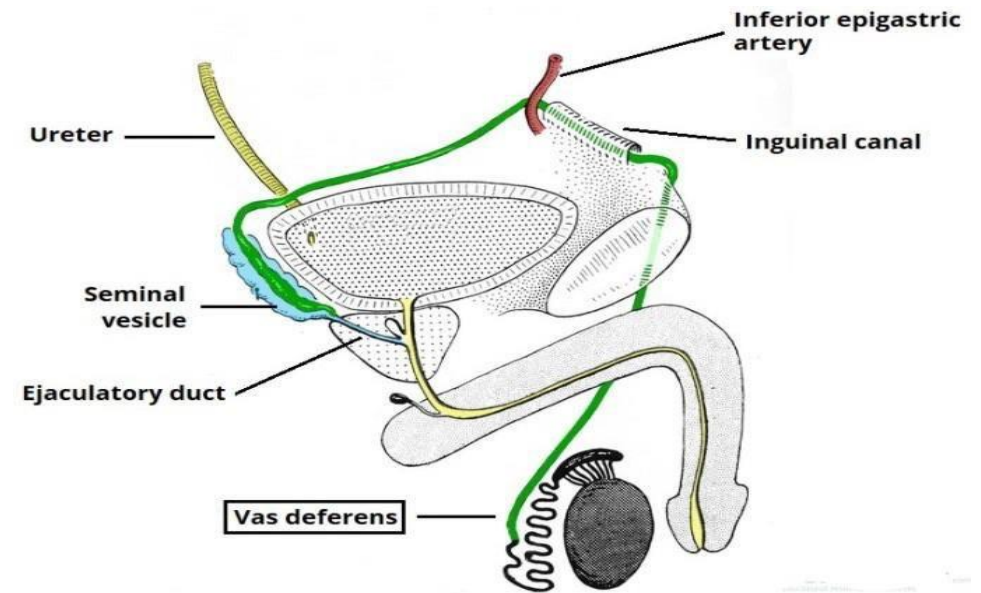
VASCULAR SUPPLY, INNERVATION AND LYMPHATIC DRAINAGE OF THE TESTES AND EPIDIDYMISS

- **Arterial Supply:**
 - Testicular arteries (which arise directly from the abdominal aorta).
 - Testes are also supplied by:
 - ▶ Cremasteric artery (from the inferior epigastric artery).
 - ▶ Artery of the vas deferens (from the inferior vesical artery).
- **Venous Drainage:**
 - Pampiniform plexus of veins which converge to form the testicular veins.
 - ▶ Left testicular vein drains into the left renal vein.
 - ▶ Right testicular vein drains into the inferior vena cava.
- **Lymphatic Drainage:**
 - To lumbar and para-aortic nodes along the lumbar vertebrae.
- **Innervation:**
 - Testicular plexus of nerves
 - ▶ Sympathetic fibers from T10-T11 segment of the spinal cord, visceral afferent fibers and parasympathetic fibers.
 - ▶ Autonomic fibers may reach the testis via the deferential plexus.



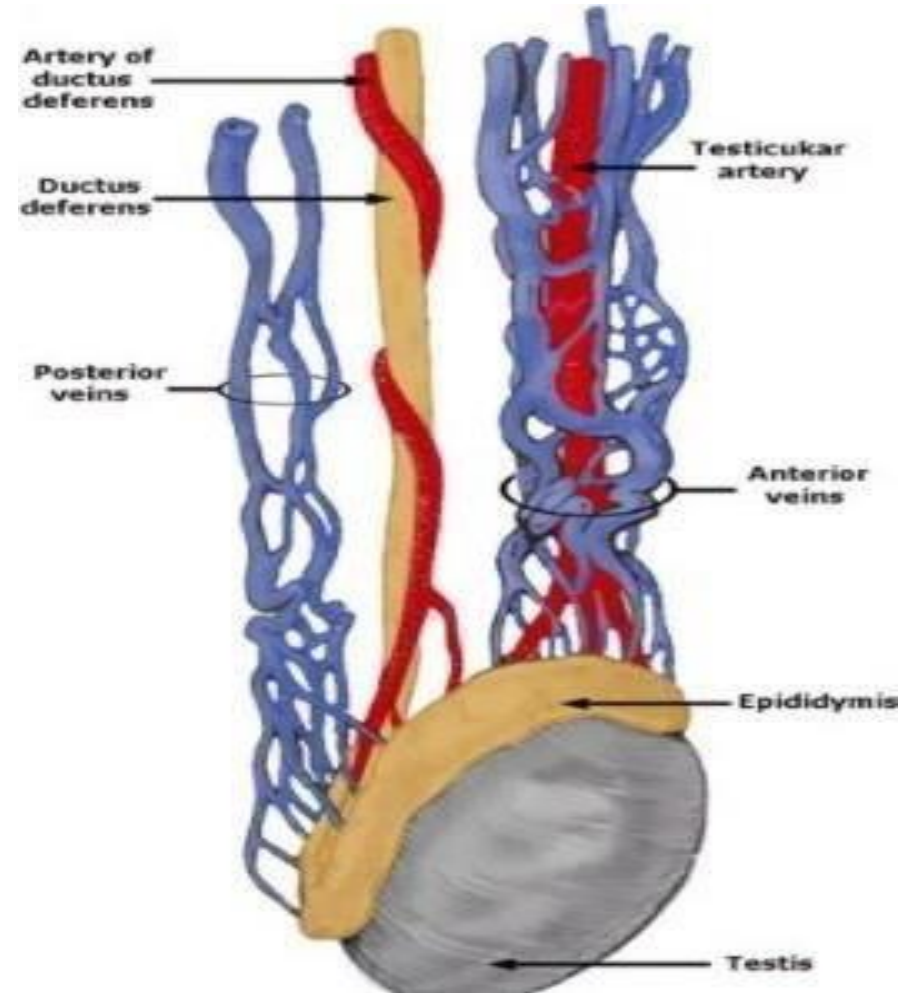
THE DUCTUS DEFERENS (VAS DEFERENS)

- Long muscular duct that transmits spermatozoa from the tail of the epididymis to the ejaculatory duct.
- The terminal portion of the vas deferens is found:
 - Posterior to the bladder.
 - Superior to the prostate.
 - Anterior to the rectum.
- The anatomical course of the vas deferens is as follows:
 - It is continuous with the tail of the epididymis.
 - Travels through the inguinal canal as part of the spermatic cord
 - Moves down the lateral pelvic wall close to the ischial spines.
 - Turns medially to pass between the bladder and the ureter and then travels downward on the posterior surface of the bladder.
 - The ductus deferens expands to form the ampulla of the ductus deferens which joins the duct from the seminal vesicle to form the ejaculatory duct.



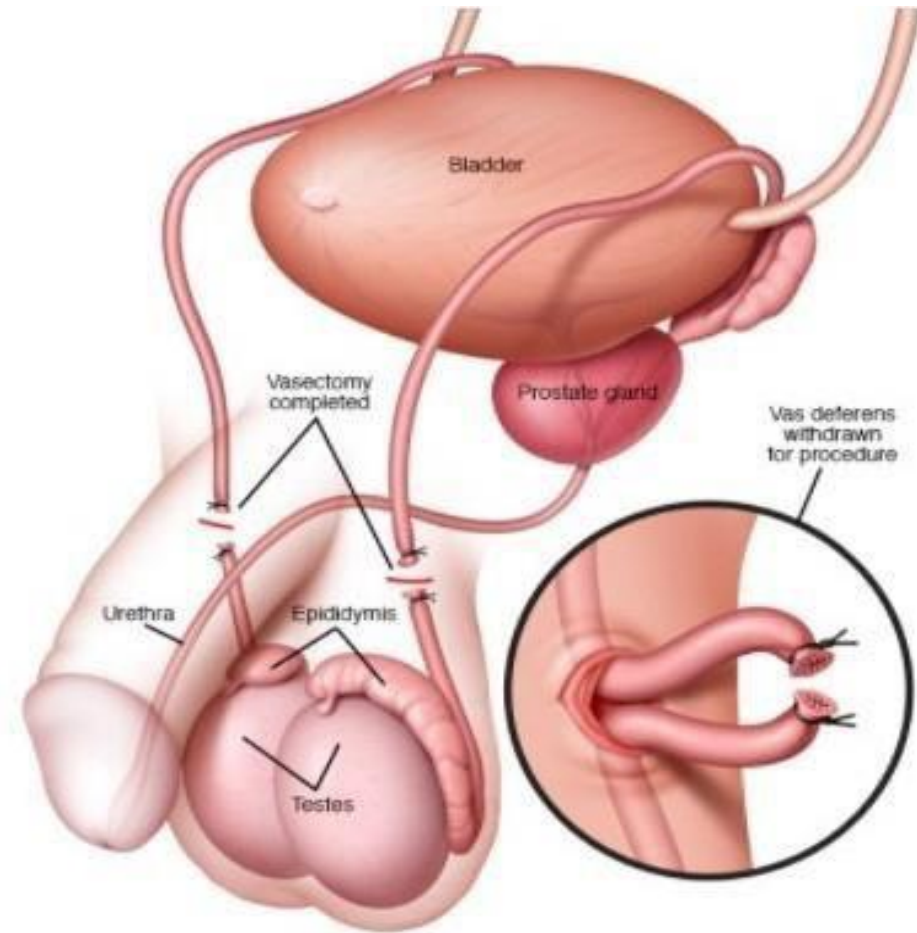
ARTERIAL SUPPLY AND VENOUS DRAINAGE OF THE DUCTUS DEFERENS

- **Arterial Supply:**
 - Artery to the ductus deferens (arise from the inferior vesical artery).
- **Venous drainage:**
 - Testicular vein and distal pampiniform plexus.
 - Terminal portion drains into the vesicular/prostatic venous plexus.



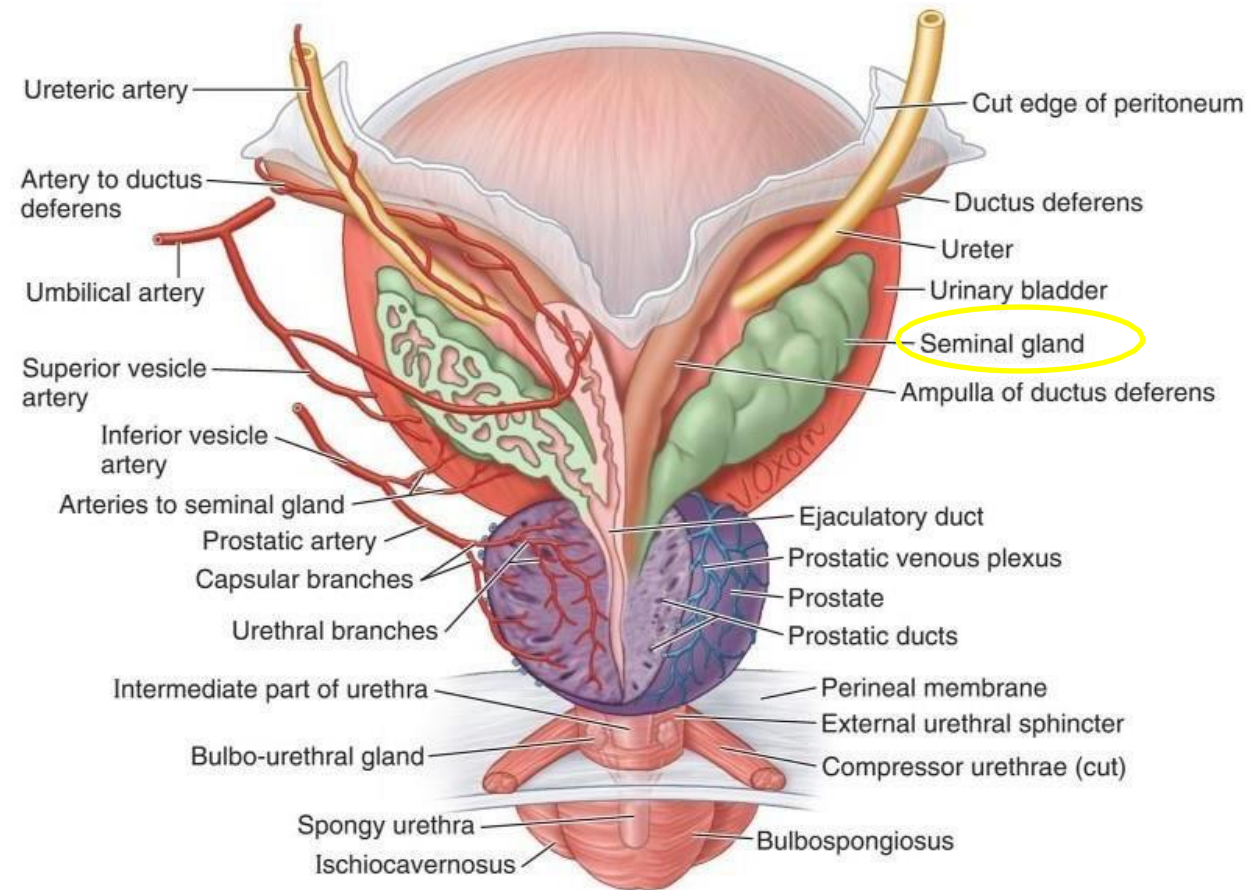
CLINICAL CORRELATE: VASECTOMY

- The ductus deferens transports sperm from the tail of the epididymis to the ejaculatory duct in the pelvic cavity.
- A vasectomy is a procedure in which the ductus deferens is surgically dissected, divided and ligated.
 - When this occurs bilaterally, the patient is sterile, so it is a useful method for male contraception.
- The ductus deferens can be easily palpated in the spermatic cord between the testes and the superficial inguinal ring and it can be accessed through the skin and superficial fascia, so hospitalization is not normally required for the procedure.



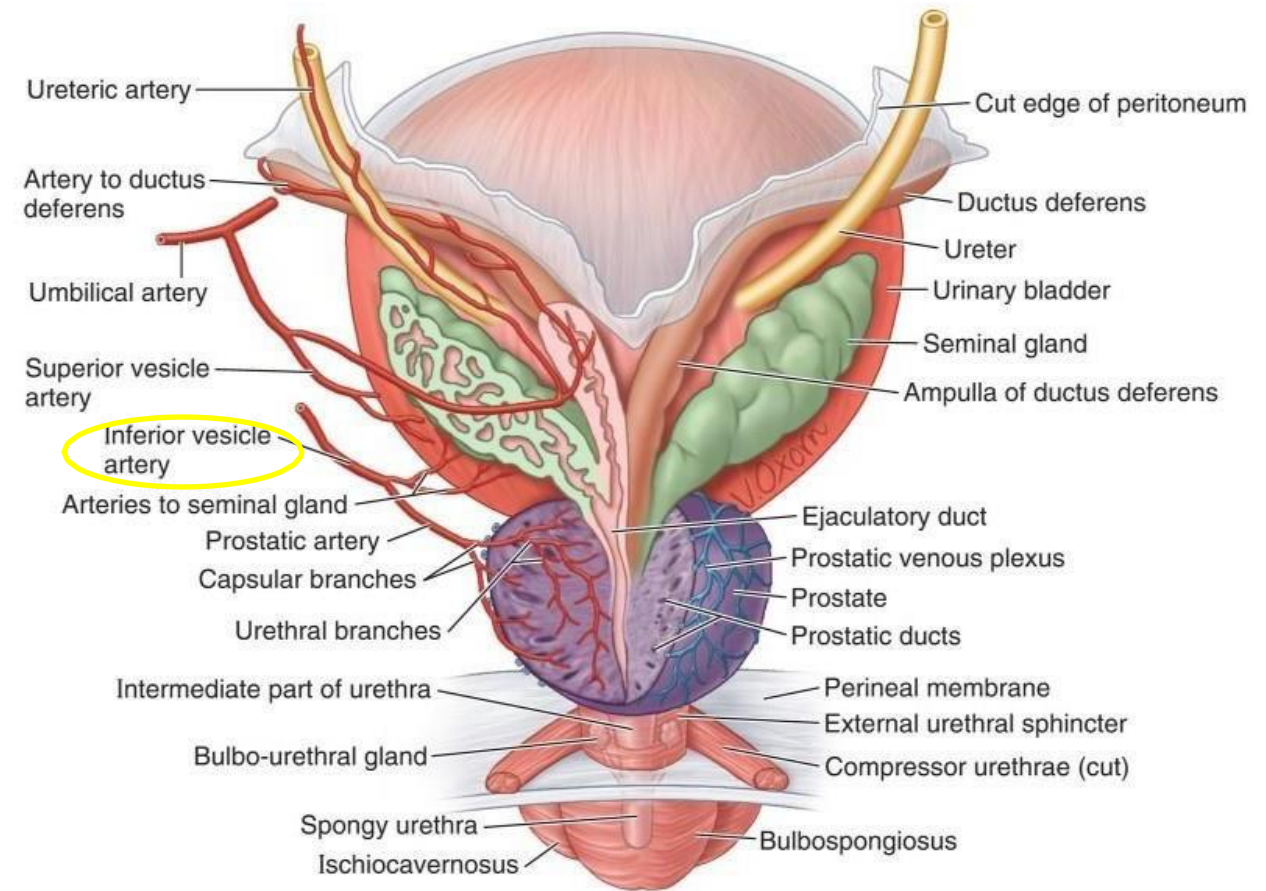
THE SEMINAL VESICLES (GLANDS)

- The seminal vesicles (glands) are elongated structures found between the fundus of the bladder and the rectum.
- **Anatomical position:**
 - Posterior to the bladder.
 - Superior to the prostate.
 - Anterior to the rectum.
- Secrete a thick alkaline fluid with fructose (energy source for sperms) and a coagulating agent that mixes with the sperms as they pass into the ejaculatory ducts and urethra. Accounts for 70% of total volume of semen.
- The duct of the seminal gland joins the ductus deferens to form the ejaculatory duct.



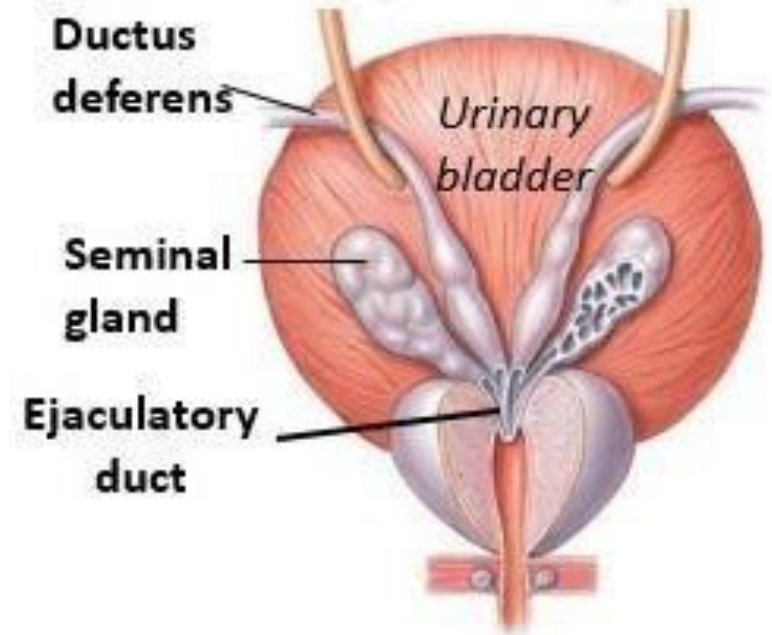
NEUROVASCULATURE AND LYMPHATIC DRAINAGE OF THE SEMINAL VESICLES

- **Arterial Supply:**
 - Inferior vesical artery
 - Internal pudendal artery
 - Middle rectal artery
- **Venous drainage:**
 - The veins accompany the arteries and have similar names.
- **Innervation:**
 - Mainly sympathetic in origin.
- **Lymphatic Drainage:**
 - External and internal iliac lymph nodes.



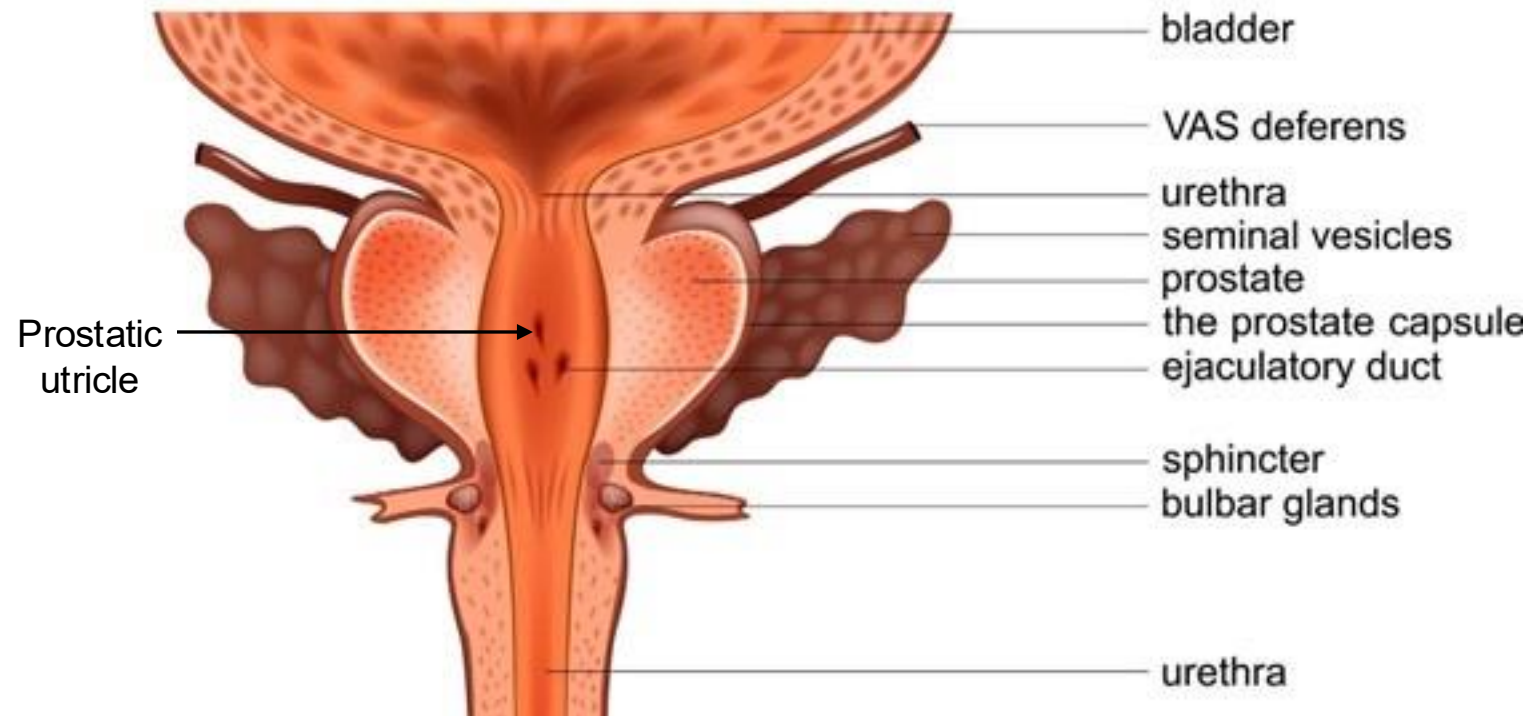
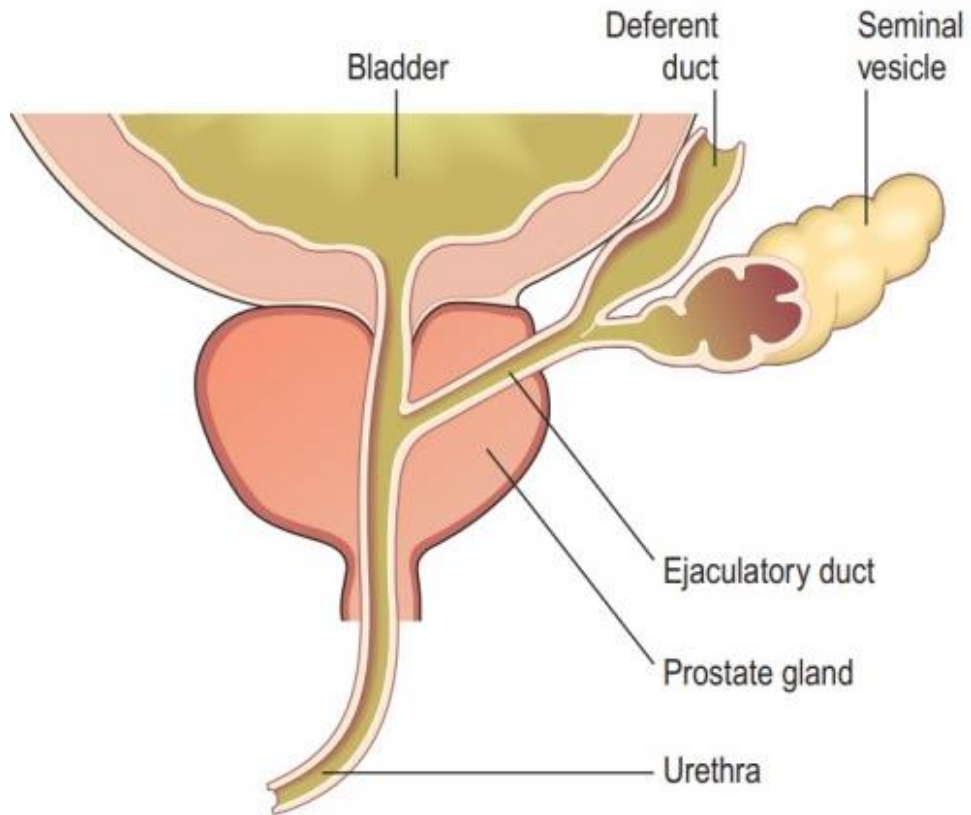
THE EJACULATORY DUCTS

- Two slender tubes that arise from the union of the seminal gland ducts and the ductus deferens.
- Arise near the neck of the bladder and run close together as they pass posteroinferiorly through the posterior part of the prostate and along the sides of the prostatic utricle.
- The ejaculatory ducts converge and open on the seminal colliculus by tiny, slit-like apertures on the opening of the prostatic utricle.
- Prostatic secretions do not join the seminal fluid until the ejaculatory ducts have terminated in the prostatic urethra.
- Deliver glandular secretions and sperm to the prostatic urethra.
- **Arterial Supply:** Arteries to the ductus deferens
- **Venous drainage:** Prostatic and venous plexuses



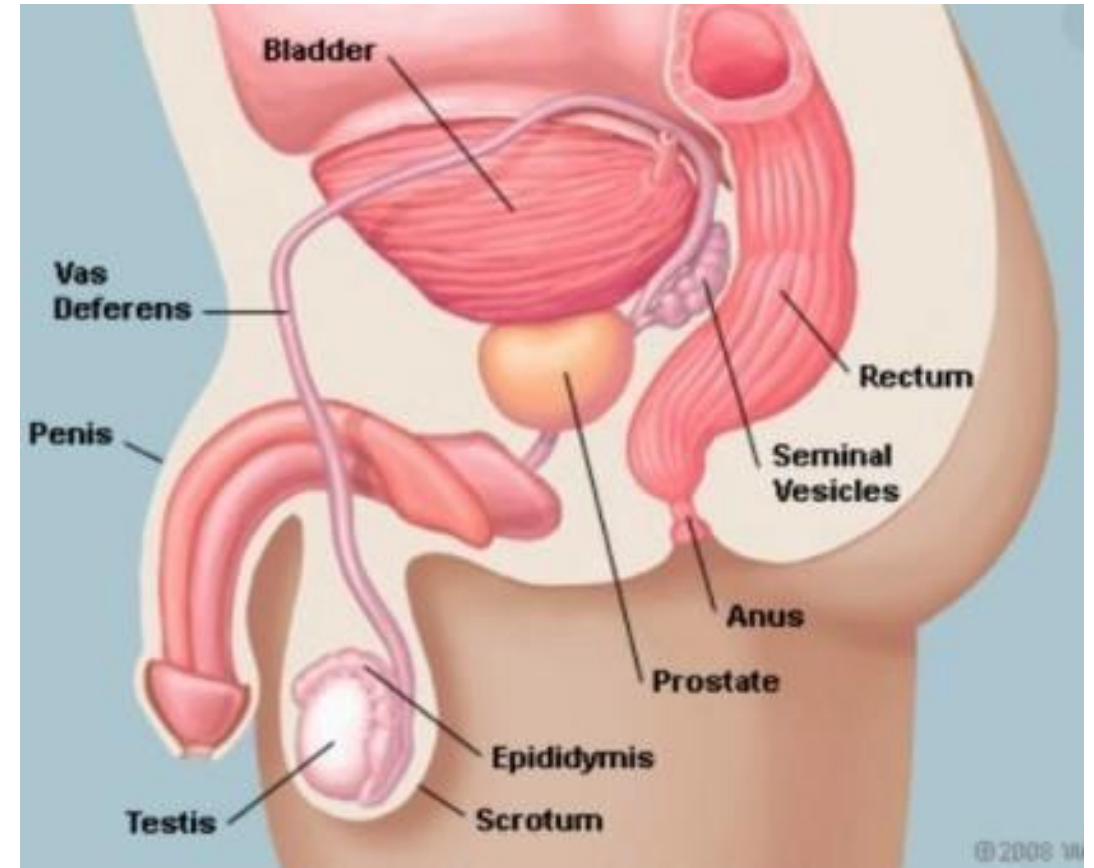
*The ejaculatory ducts are located inside the prostate gland tissue in the posterior aspect

THE EJACULATORY DUCT



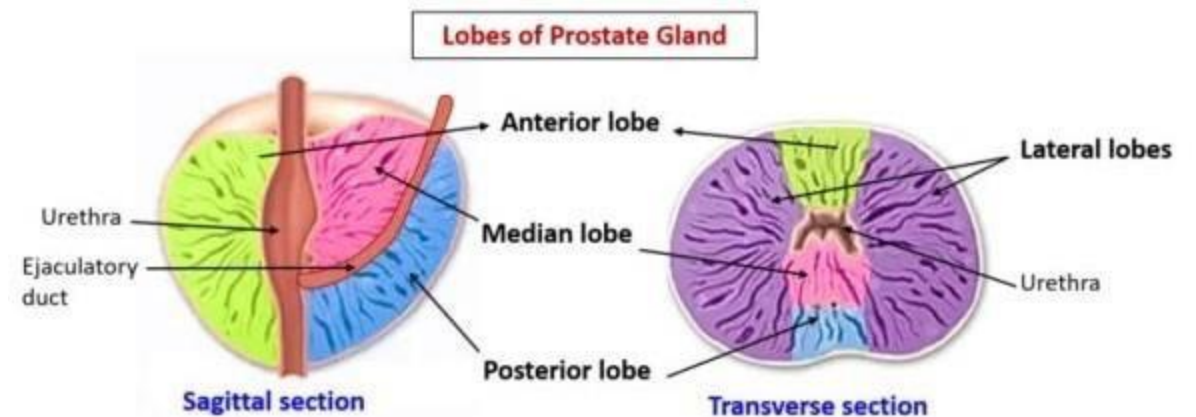
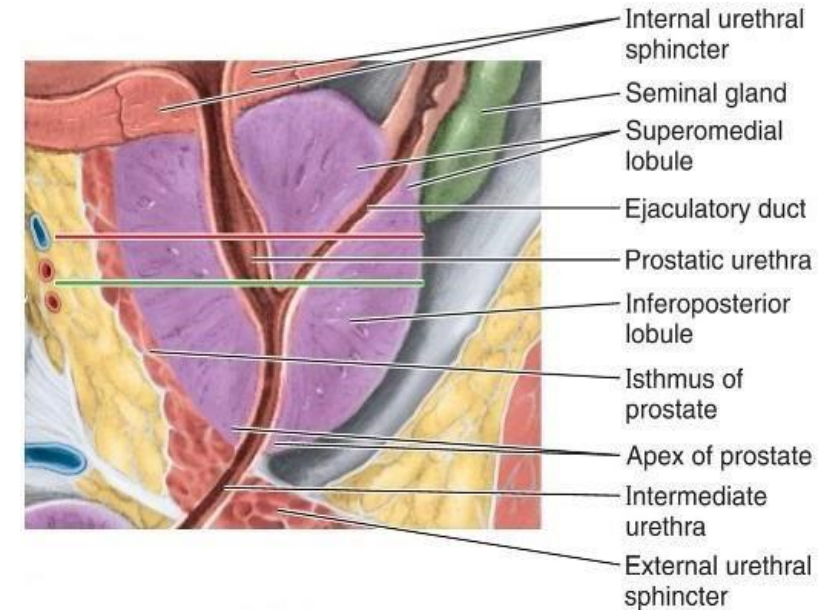
THE PROSTATE

- Fibromuscular glandular organ (2/3 glandular; 1/3 fibromuscular).
- Secretes proteolytic enzymes into the semen which acts to break down clotting factors in the ejaculate. This allows the semen to remain in a fluid state, moving throughout the female reproductive tract for potential fertilization.
- Anatomical position:
 - **Inferior** to the neck of the bladder
 - **Superior** to the external urethral sphincter.
 - **Anterior** to the ampulla of the rectum.
 - **Posterior** to the pubic symphysis.
 - Prostate surrounds the prostatic urethra.



THE PROSTATE

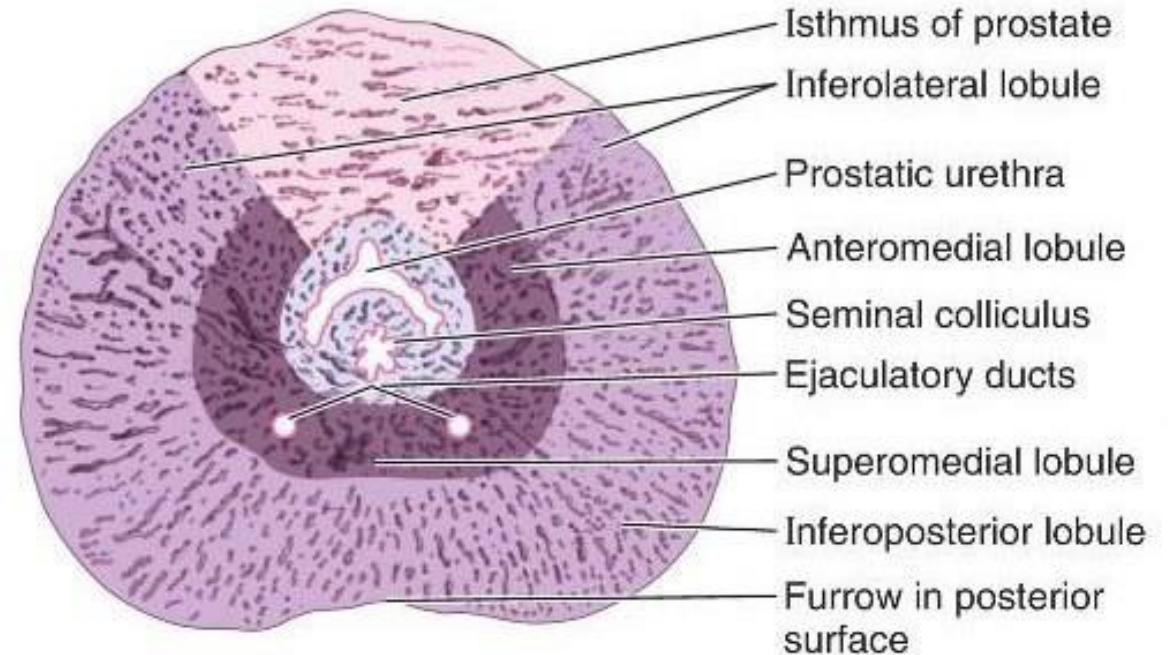
- Prostate is surrounded by a thin **fibrous capsule of the prostate**.
- The prostate has:
 - A base closely related to the neck of the bladder.
 - An apex
 - Anterior surface separated from the pubic symphysis by retroperitoneal fat in the retropubic space.
 - Posterior surface that is related to the ampulla of the rectum.
 - Inferolateral surfaces related to the levator ani muscle.
 - The isthmus of the prostate
 - 5 lobes of the prostate:
 - ▶ Anterior lobe
 - ▶ Median lobe
 - ▶ Lateral (right and left) lobes
 - ▶ Posterior lobe



THE PROSTATE

- The lateral (right and left) lobes of the prostate are further subdivided into **anatomical lobules** by the urethra and the ejaculatory ducts as they pass through the organ. These anatomical lobules are:
 - **Inferoposterior lobule:**
 - ▶ Lies posterior to the urethra and inferior to the ejaculatory ducts.
 - **Inferolateral lobule:**
 - ▶ Lies lateral to the urethra
 - ▶ Forms the major part of the right or left lobe.
 - **Superomedial lobule:**
 - ▶ Surrounds the ejaculatory duct.
 - **Anteromedial lobule:**
 - ▶ Lies directly lateral to the proximal prostatic urethra

Transverse section of the prostate



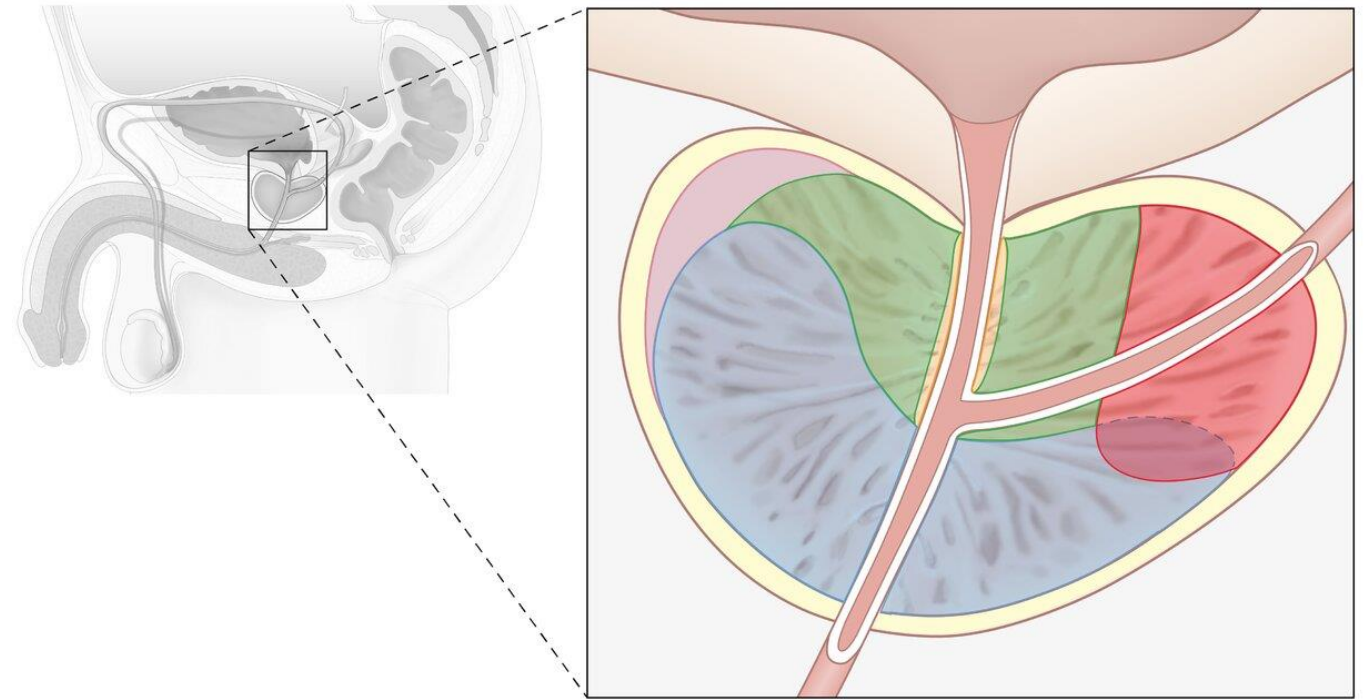
THE PROSTATE

■ **Transitional zone:**

Most common site for hyperplasia (BPH)

■ **Peripheral zone:**

The most common part of the prostate for prostate cancer to develop.

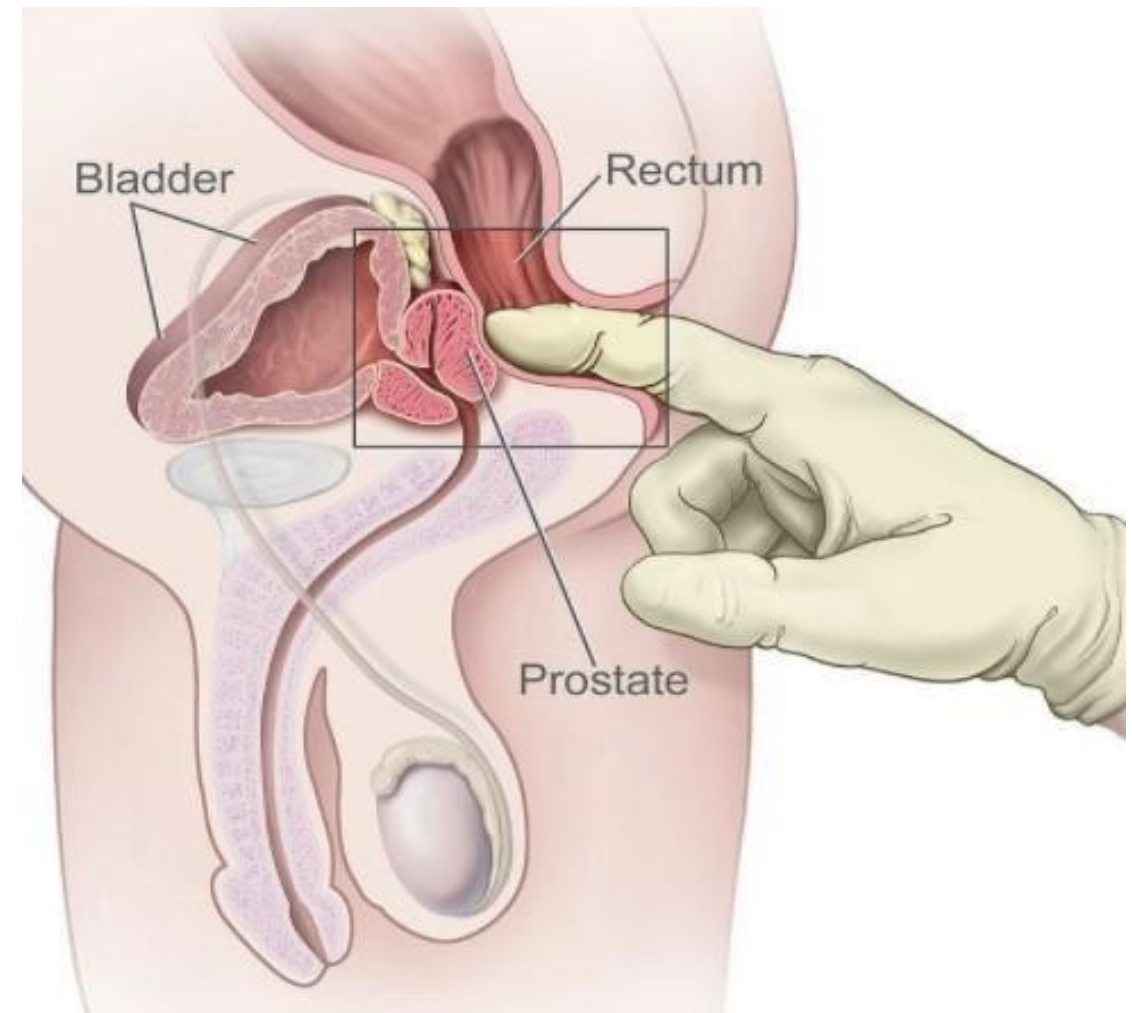


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|--------------------|---------------------|
| 1 Prostate capsule | 4 Periurethral zone |
| 2 Anterior zone | 5 Central zone |
| 3 Transition zone | 6 Peripheral zone |

- **Arterial Supply:**
 - Prostatic arteries (branches of the internal iliac arteries).
- **Venous Drainage:**
 - Prostatic venous plexus (drains into the internal iliac veins).
- **Lymphatic Drainage:**
 - Obturator and internal iliac lymph nodes
- **Innervation:**
 - Inferior hypogastric plexus (receives sympathetic, parasympathetic and sensory innervation).
 - Smooth muscle of the prostate gland is innervated by sympathetic fibers which activate during ejaculation.

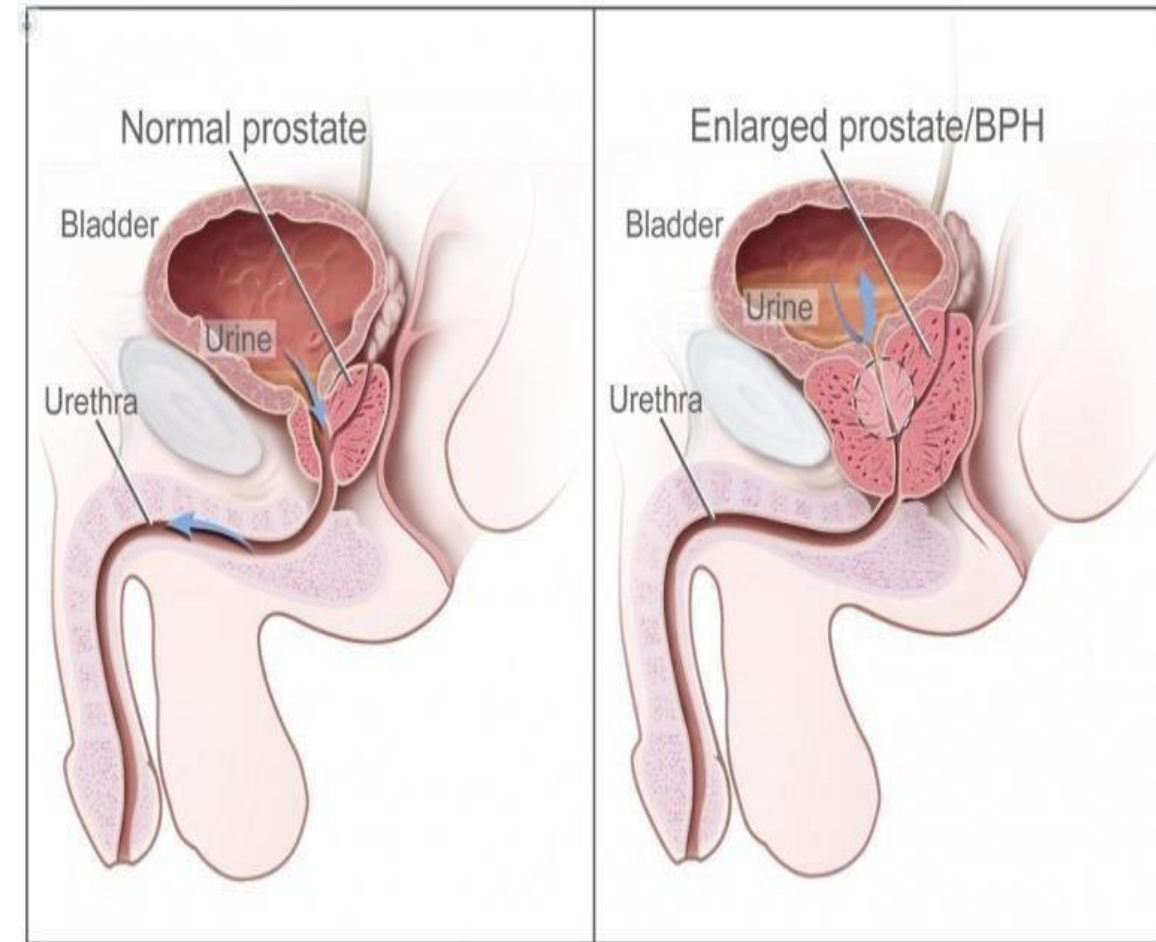
CLINICAL CORRELATE: DIGITAL RECTAL EXAMINATION

- Examination of the lower portion of the rectum, anus, prostate gland and the perineum.
- During the examination, the examining finger can also palpate the prostate gland.
- A digital rectal examination is indicated as part of a full physical examination.
- Disease processes that may be investigated with a digital rectal examination include:
 - Benign prostatic hypertrophy
 - Prostate cancer
 - Hemorrhoids
 - Anal and rectal cancers
 - Anal fissures
 - Constipation



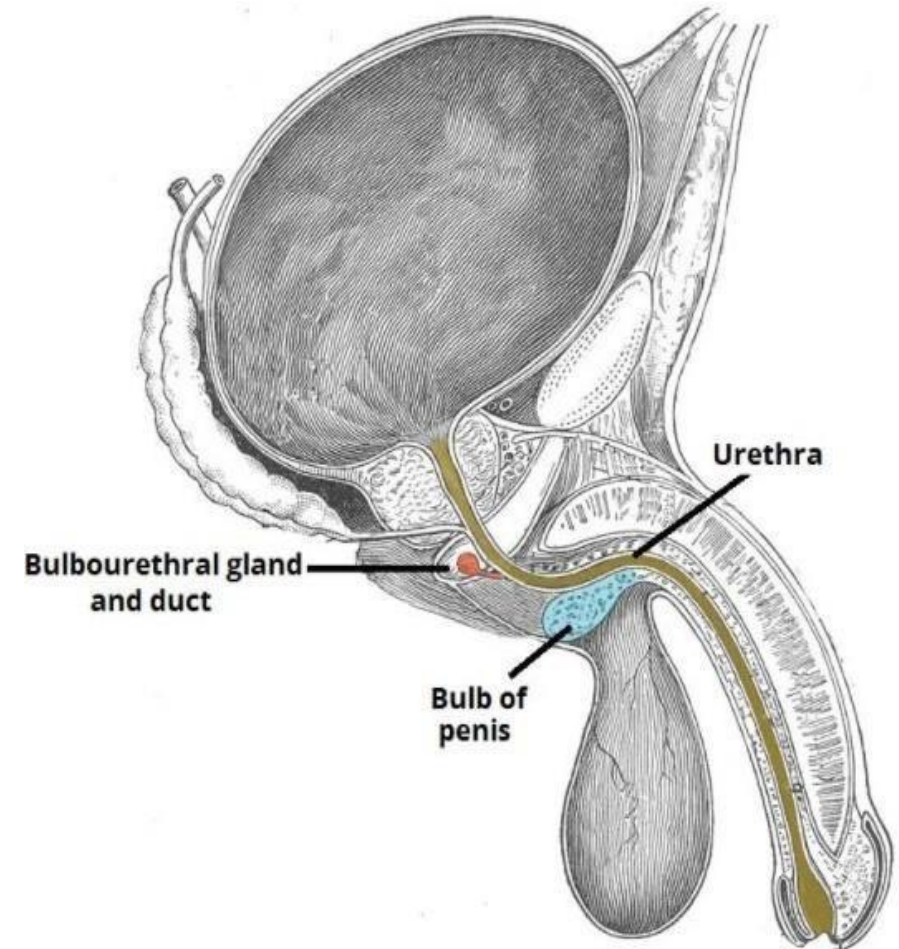
CLINICAL CORRELATE: BENIGN PROSTATIC HYPERTROPHY

- Most older males suffer from benign prostatic hypertrophy or enlargement of the prostate.
- An enlarged prostate projects into the urinary bladder and compresses the prostatic urethra leading to difficulties with urination.
- The prostate feels “bulky” on digital rectal examination.
- Usually caused by hyperplasia of the glands in the middle lobe or the transitional zone of the prostate.
- Symptoms include:
 - Nocturia (need to void during the night)
 - Dysuria (difficulty and/or pain during urination).
 - Urgency (sudden desire to void).
 - Difficulty initiation micturition, poor stream and terminal dribbling of urine.



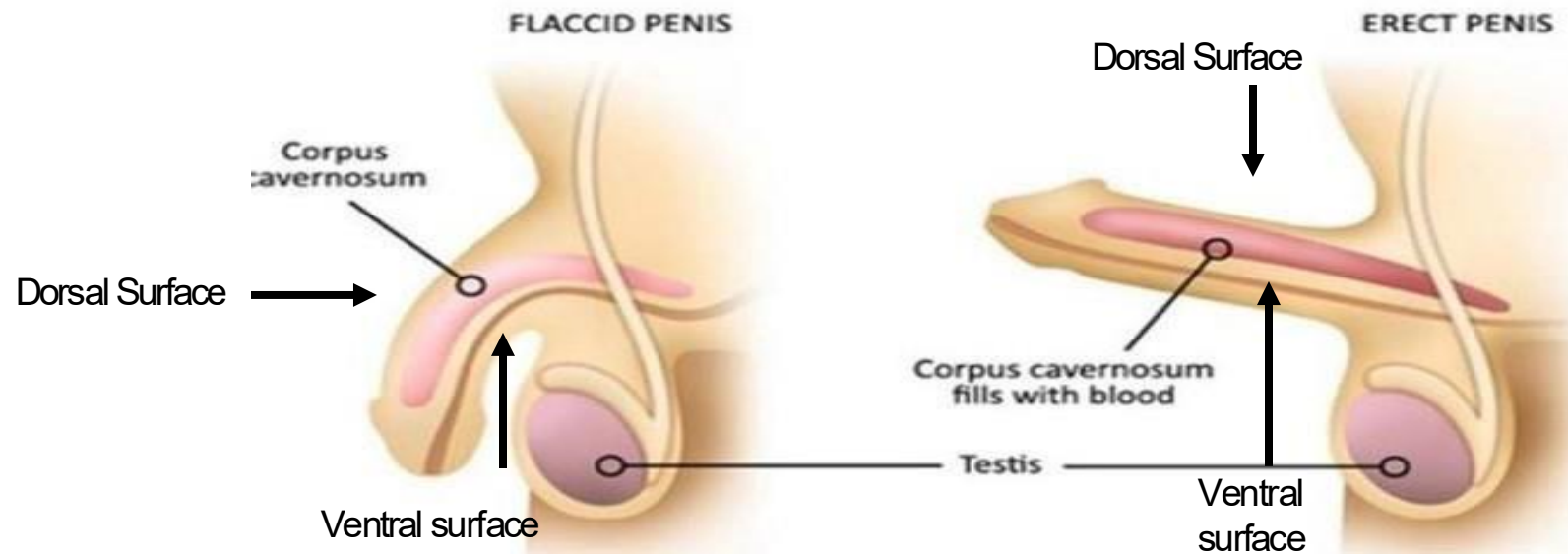
BULBO-URETHRAL GLANDS (COWPER'S GLANDS)

- Small, pea-shaped mucus glands situated within the deep perineal pouch, posterolateral to the membranous urethra.
- The duct from each gland passes through the perineal membrane to open into the bulb of the spongy urethra at the root of the penis.
- Contribute to lubrication of the urethra and the total volume of semen by producing a lubricating mucus secretion.
- **Arterial Supply:** Arteries to the bulb of the penis.
- **Lymphatic Drainage:** Internal and external iliac lymph nodes.
- **Innervation:** Hypogastric nerve, pelvic nerve or pelvic branch of the pudendal nerve.



THE PENIS

- The penis is the external organ of the male reproductive system with two main functions:
 - Sexual intercourse
 - Micturition
- In the anatomical position, the penis is erect.
- When the penis is flaccid, the dorsum of the penis is directed anteriorly.



THE PENIS

- The penis can be divided into three parts:

- **Root**

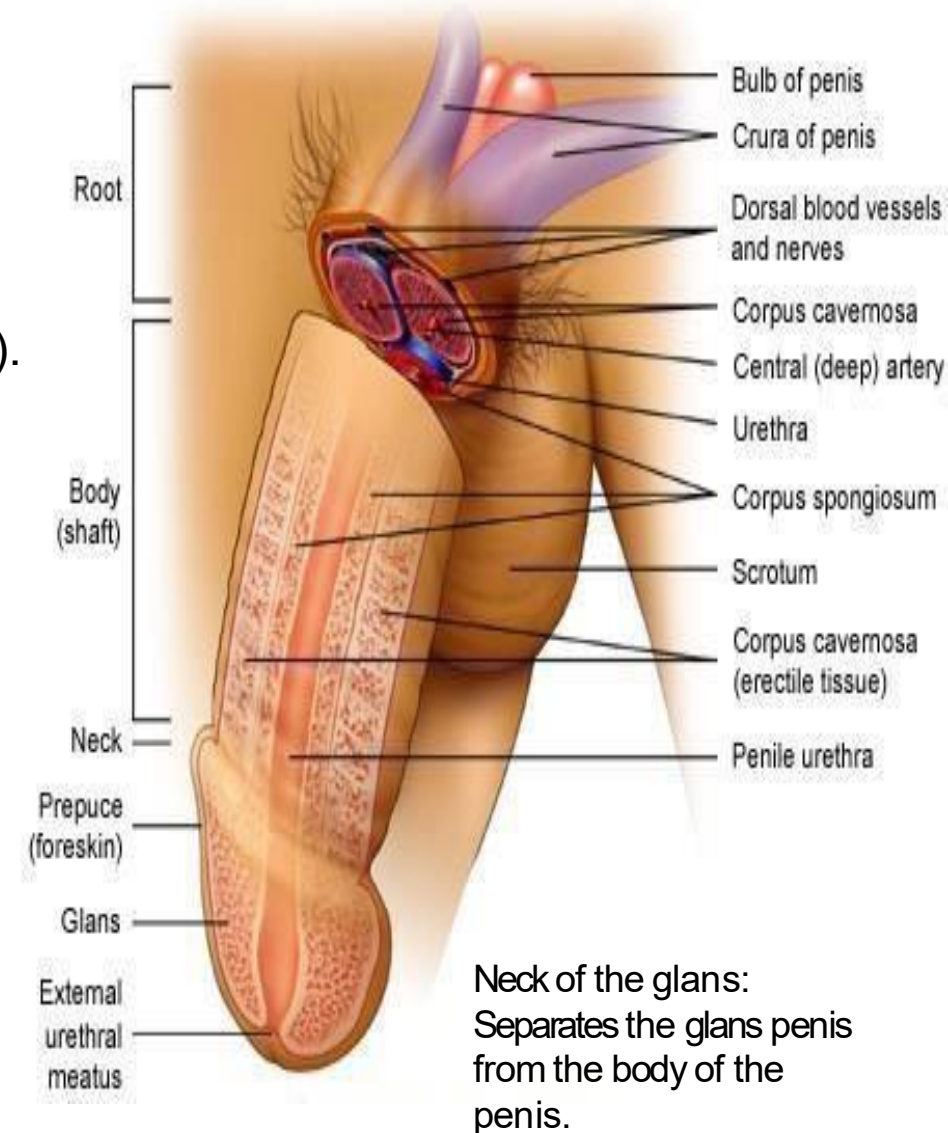
- ▶ Most proximal part of the penis
- ▶ Located in the superficial perineal pouch of the pelvic floor.
- ▶ Contains three erectile tissues (2 crura and bulb of the penis).
- ▶ Contains two muscles (ischiocavernosus and bulbospongiosus).

- **Body**

- ▶ Free pendulous part of the penis
- ▶ Suspended from the pubic symphysis.
- ▶ Composed of three cylinders of erectile tissue (two corpora cavernosa and the corpus spongiosum).

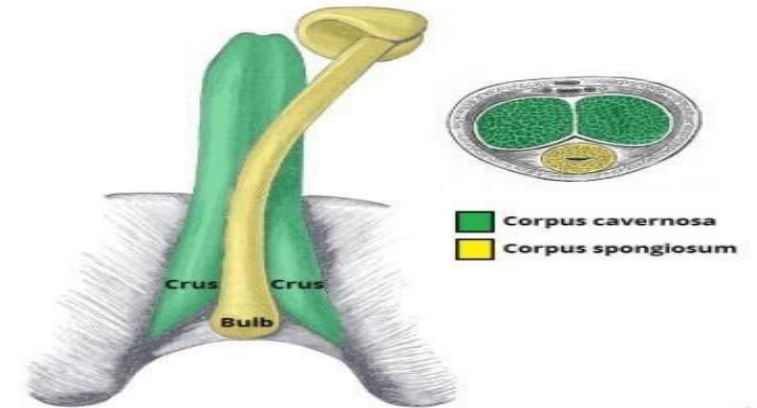
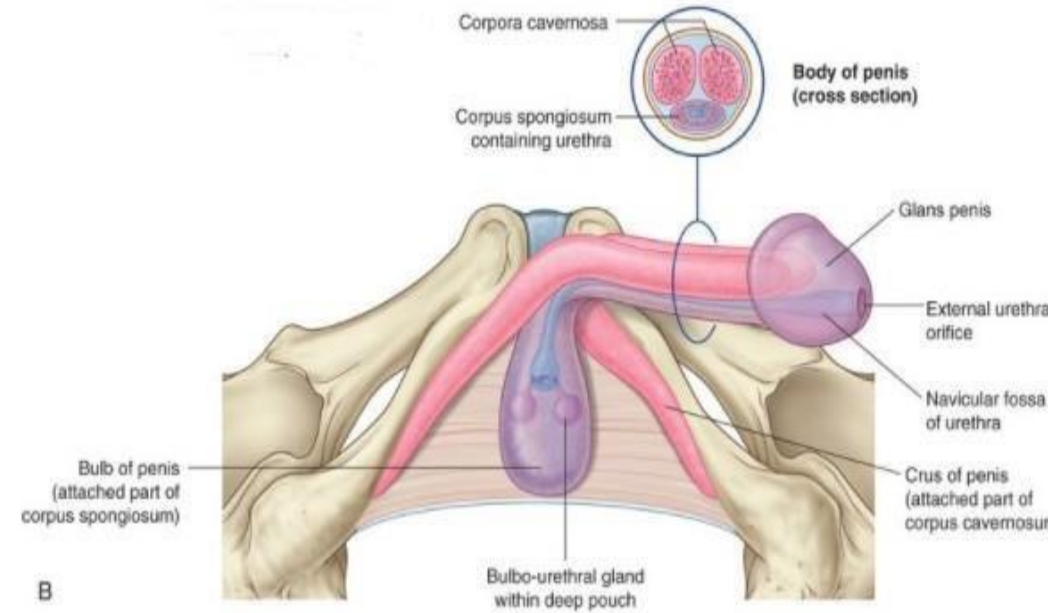
- **Glans penis**

- ▶ Most distal part of the penis.
- ▶ Conical in shape and formed by the distal expansion of the corpus spongiosum.
- ▶ Contains the opening of the urethra, termed the external urethral orifice.



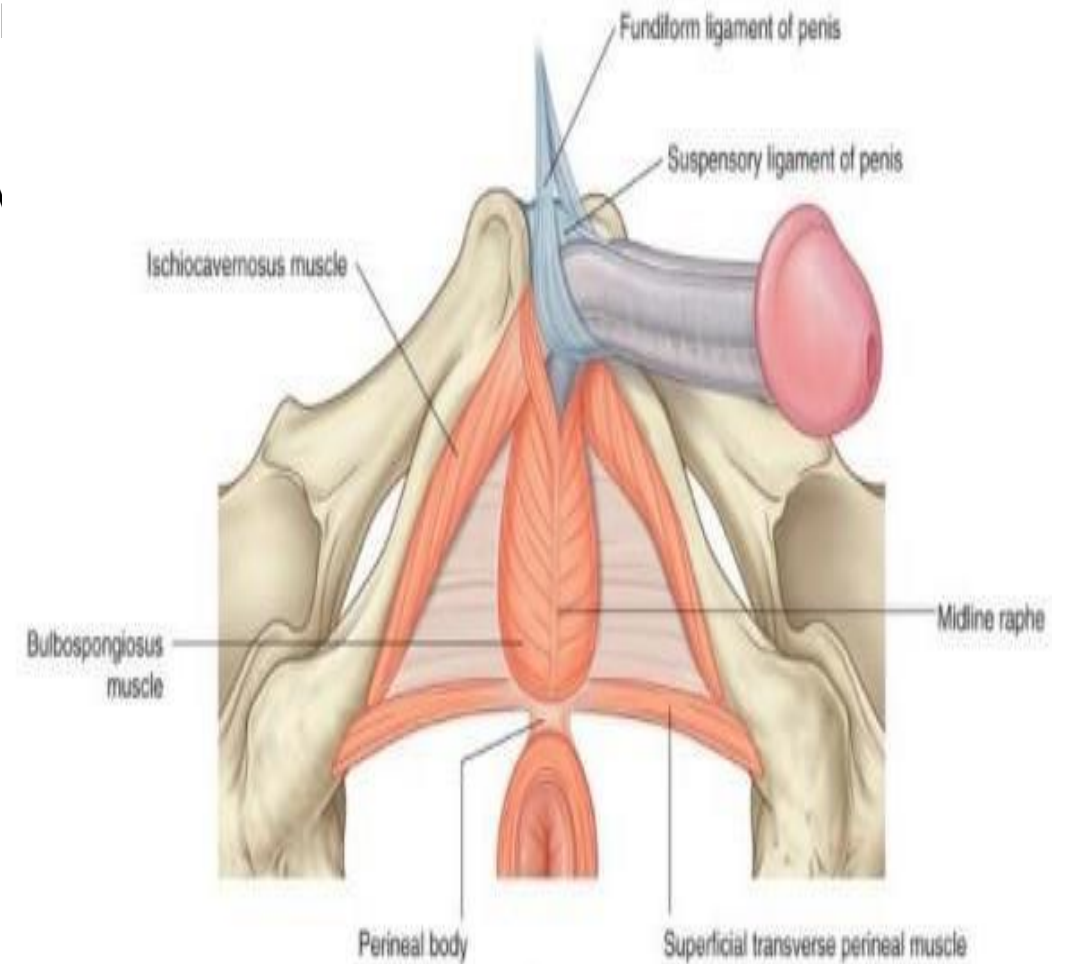
THE PENIS

- The root and body of the penis contain three masses of erectile tissue which fill with the blood during sexual arousal.
 - **Left and Right Crura (→ Corpora cavernosa)**
 - ▶ Found in the root of the penis.
 - ▶ Located laterally and attach to the ipsilateral ischial ramus and covered by the ischiocavernosus muscles.
 - ▶ Continue into the body of the penis as the 2 **corpora cavernosa**
 - **Bulb of the Penis (→ Corpus Spongiosum)**
 - ▶ Found in the root of the penis.
 - ▶ Situated in the midline of the penile root and traversed by the urethra.
 - ▶ Forms the corpus spongiosum in the body of the penis.



THE PENIS

- The muscles located in the root of the penis:
 - **Ischiocavernosus Muscle (x2)**
 - ▶ Surrounds the left and right crura of the penis in the root of the penis.
 - ▶ Contracts to force blood from the cavernous spaces in the crura into the corpus cavernosus. It compresses the deep dorsal vein of the penis, reducing venous return.
 - ▶ Helps maintain an erection.
 - **Bulbospongiosus Muscle (x2)**
 - ▶ Associated with the bulb of the penis.
 - ▶ Contracts to empty the spongy urethra of any residual semen and urine.
 - ▶ Anterior fibers aid in maintaining an erection by increasing the pressure in the bulb of the penis and impeding venous return from the penis and maintaining erection.
 - **Superficial Transverse Perineal Muscle**
 - ▶ Stabilizes the perineal body.



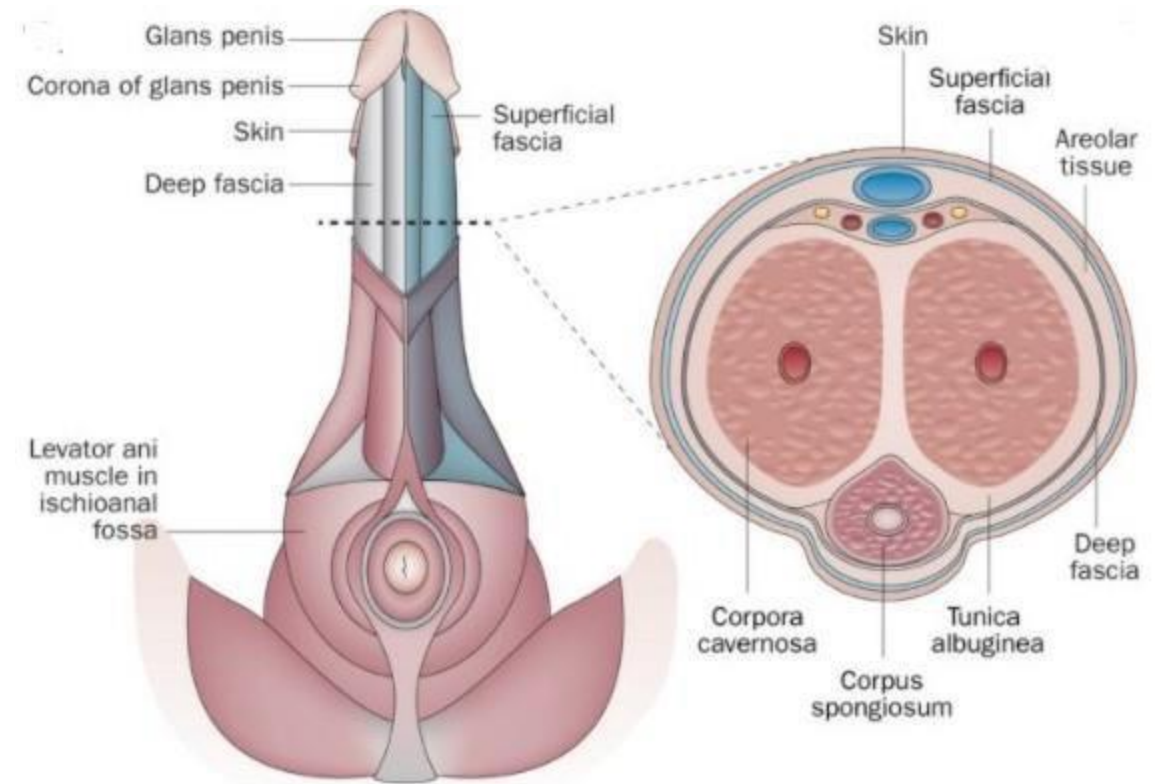
THE PENIS

Muscles	Origin	Insertion	Innervation	Function
Ischiocavernosus	Ischial tuberosity and ramus	Crus of penis and clitoris	Pudendal nerve (S2 to S4)	Move blood from crura into the body of the erect penis and clitoris
Bulbospongiosus	In women: perineal body In men: perineal body, midline raphe	In women: bulb of vestibule, perineal membrane, body of clitoris, and corpus cavernosum In men: bulbospongiosus, perineal membrane, corpus cavernosum	Pudendal nerve (S2 to S4)	Move blood from attached parts of the clitoris and penis into the glans In men: removal of residual urine from urethra after urination; pulsatile emission of semen during ejaculation
Superficial transverse perineal	Ischial tuberosity and ramus	Perineal body	Pudendal nerve (S2 to S4)	Stabilize the perineal body



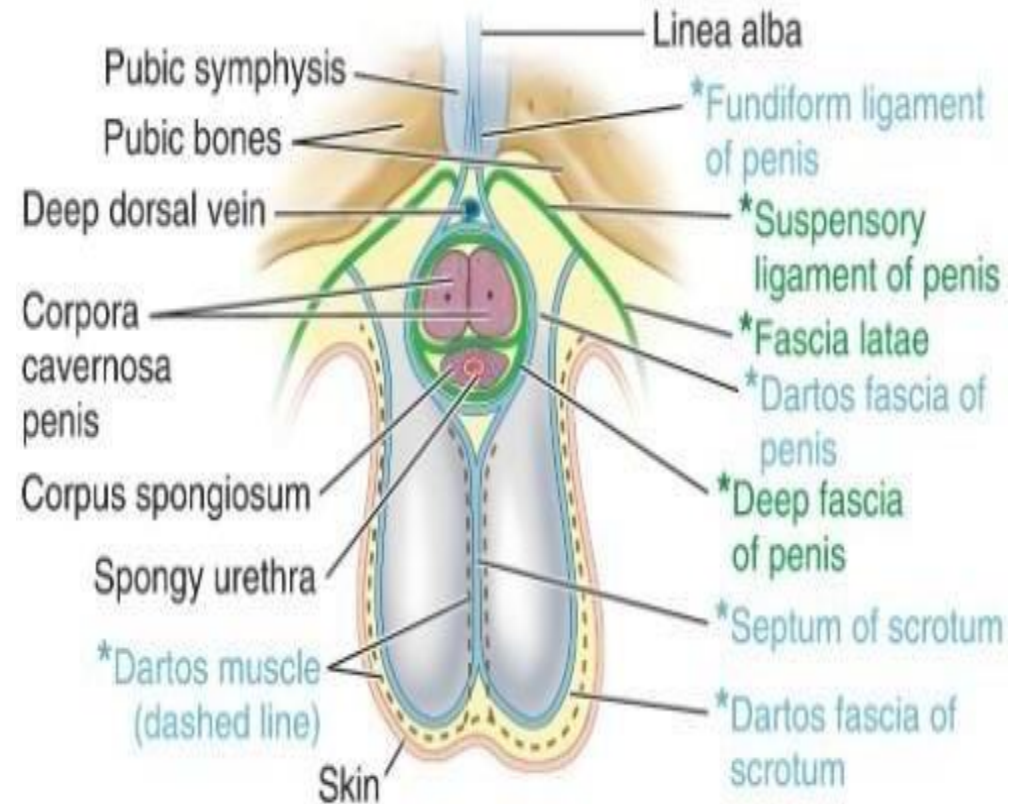
THE PENIS

- Fascial Coverings:
 - Each mass of erectile tissue has two fascial coverings:
 - ▶ **External fascia of Colles:** Most superficial layer immediately under the skin (it is in continuity with the fascia of the Scarpa)
 - ▶ **Deep fascia of the penis (Buck's fascia):** Continuation of the deep perineal fascia and forms a strong membranous covering with holds all three erectile tissues together).
 - ▶ **Tunica albuginea:** Strong fascia which forms an individual capsule around each cavernous body and is fused in the midline. The incomplete septum between the two corpora is comprised of the tunica albuginea.



THE PENIS

- The root of the penis is supported by 2 ligaments which attach it to surrounding structures:
 - **Suspensory Ligament of the penis:**
 - ▶ Condensation of deep fascia which connects the erectile bodies of the penis to the pubic symphysis.
 - **Fundiform Ligament of the penis:**
 - ▶ Irregular mass of collagen and elastic fibers of the subcutaneous tissue which runs down from the linea alba, surrounding the penis like a sling and at the level of the pubic bone and unites inferiorly at the septum of the scrotum.



THE PENIS

- The skin of the penis is thin and more heavily pigmented than the rest of the body.
- Skin is connected to the underlying fascia by loose connective tissue.
- The prepuce (foreskin) is a double layer of skin and fascia found at the neck of the glans, which covers the glans penis to a variable extent.
- The prepuce is connected to the surface of the glans by the **frenulum of the prepuce**, a median fold that passes from the deep layer of the prepuce to the urethral surface of the glans.

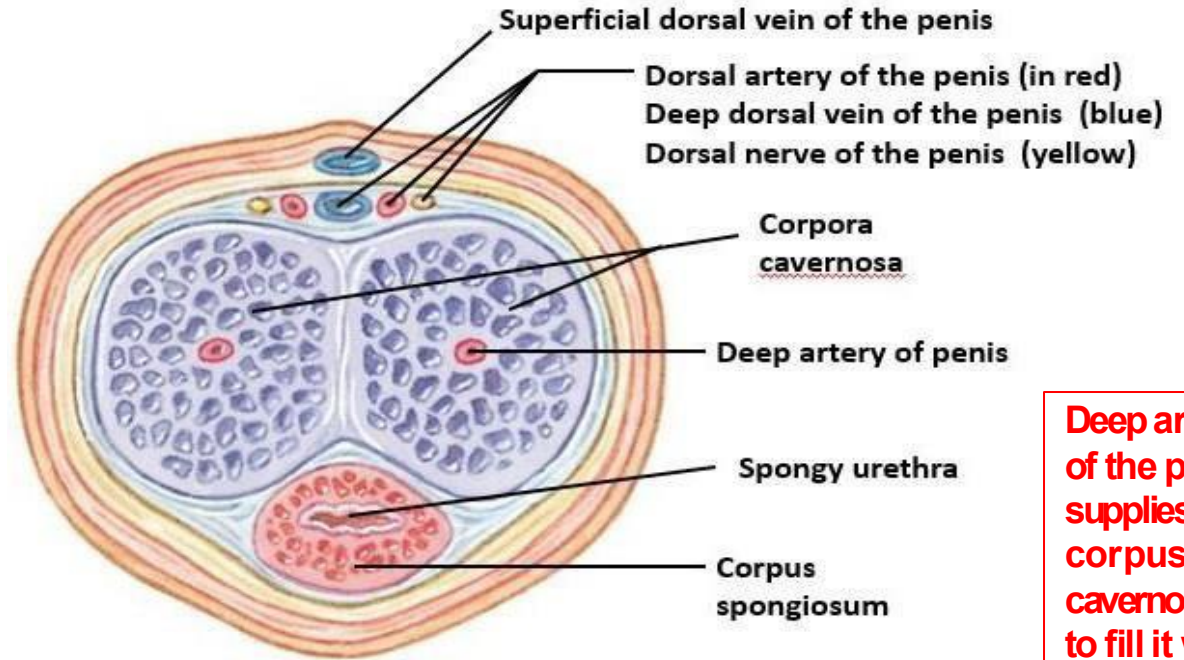


NEUROVASCULAR SUPPLY TO THE PENIS

- **Arterial Supply:**

- Dorsal arteries of the penis
- Deep arteries of the penis
- Bulbourethral artery

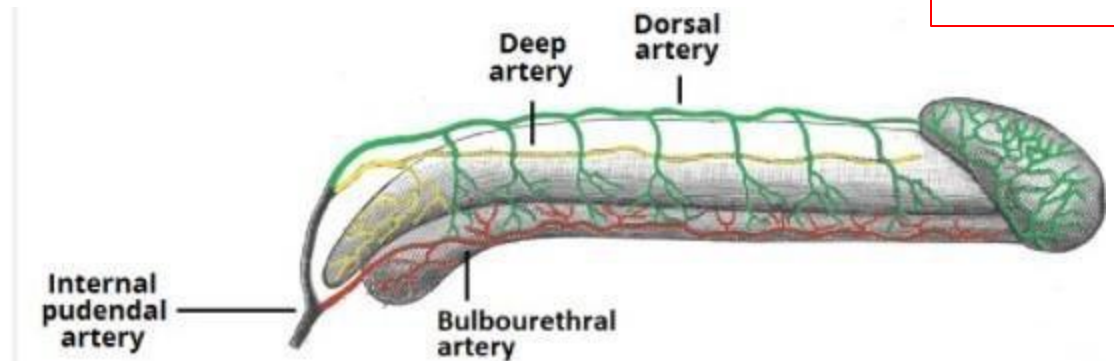
*These are all branches of the **internal pudendal artery** which is a branch of the anterior division of the **internal iliac artery**.



Deep artery of the penis supplies the corpus cavernosum to fill it with blood for an erection.

- **Venous drainage:**

- Cavernosus spaces are drained by the deep dorsal vein of the penis which empties into the prostatic plexus.
- Superficial dorsal vein drains the superficial structures of the penis such as the skin and cutaneous tissues.



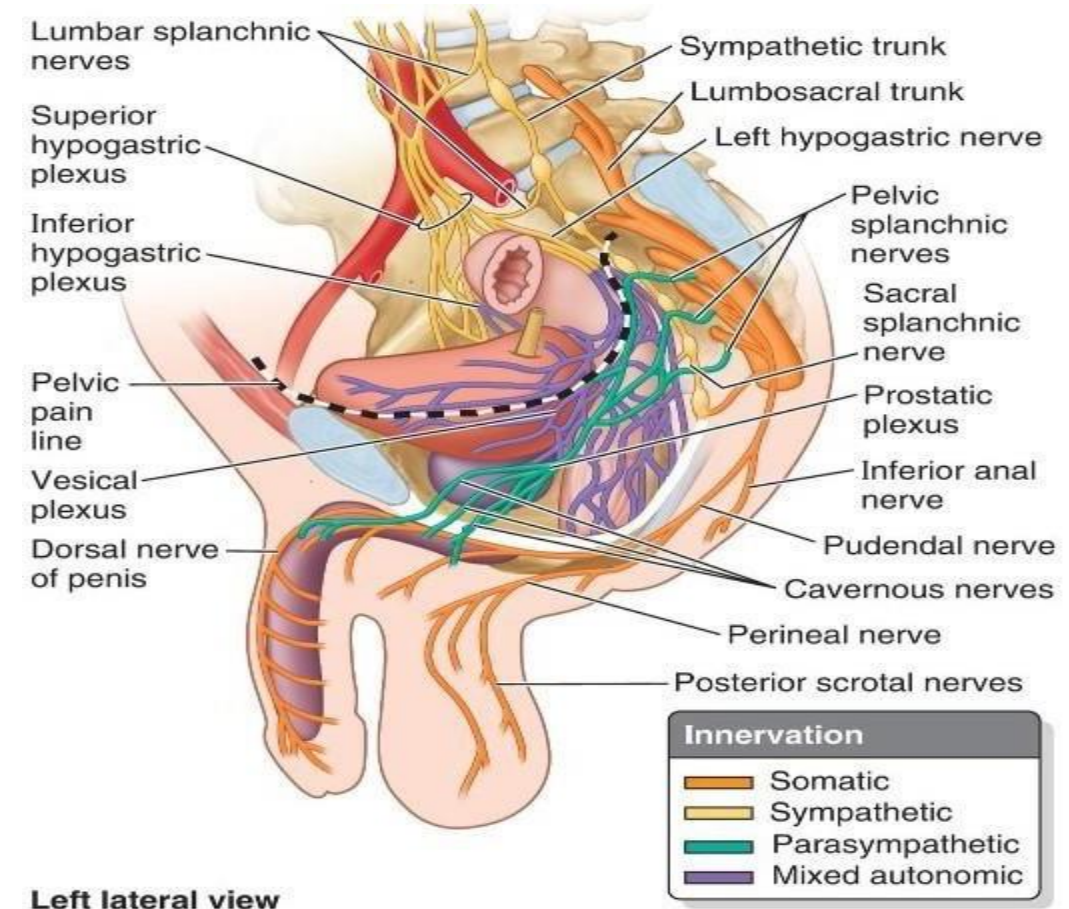
NEUROVASCULAR SUPPLY TO THE PENIS

- **Innervation**

- Penis is supplied by S2-S4 spinal cord segments and spinal ganglia.
- Sensory and sympathetic innervation to the skin and glans penis is supplied by the **dorsal nerve of the penis** (branch of the pudendal nerve).
- Parasympathetic innervation is carried by the cavernous nerves from the peri-prostatic nerve plexus and is responsible for the vascular changes which causes erection.

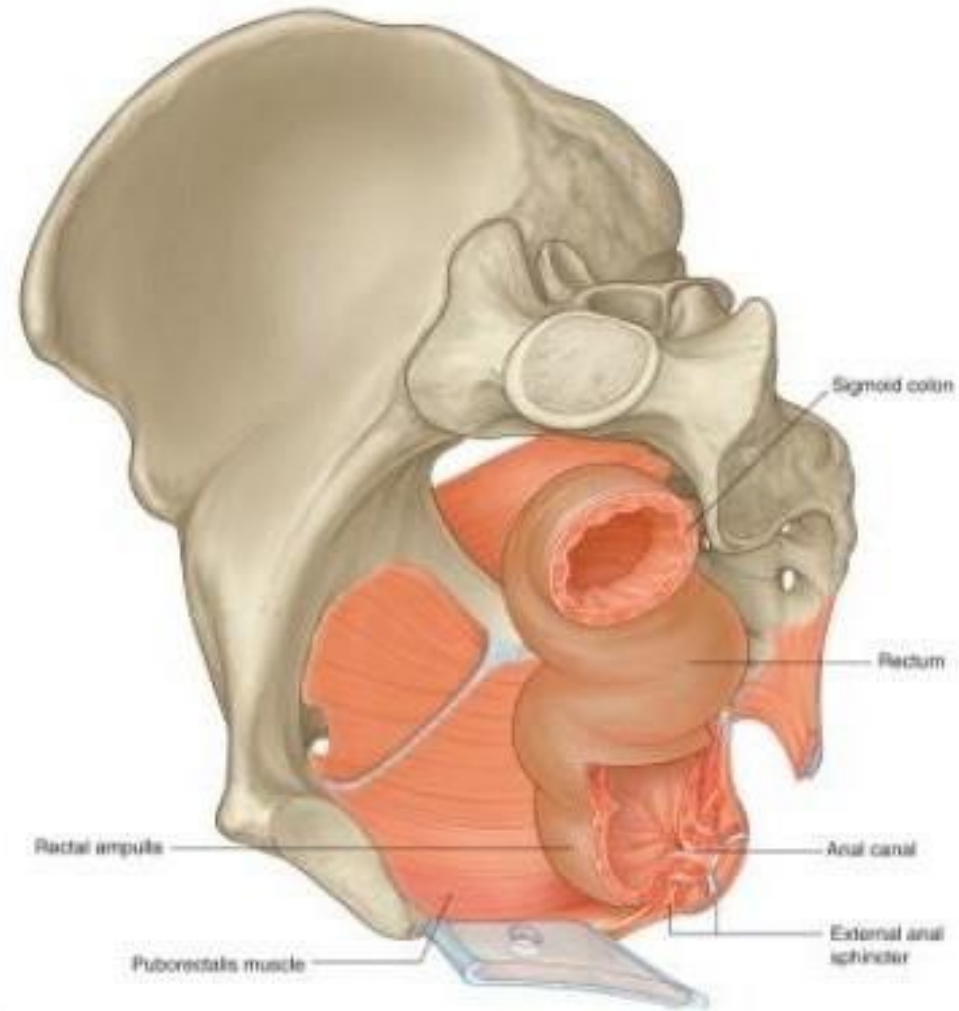
- **Nerve Control of Erection and Ejaculation in a Male:**

- **Erection:** Parasympathetic nerve impulses cause blood to accumulate in erectile tissues.
- **Emission and Ejaculation:** Sympathetic nerve impulses.
 - ▶ Emission: Movement of semen into the urethra.
 - ▶ Ejaculation: Movement of semen out of the urethra.



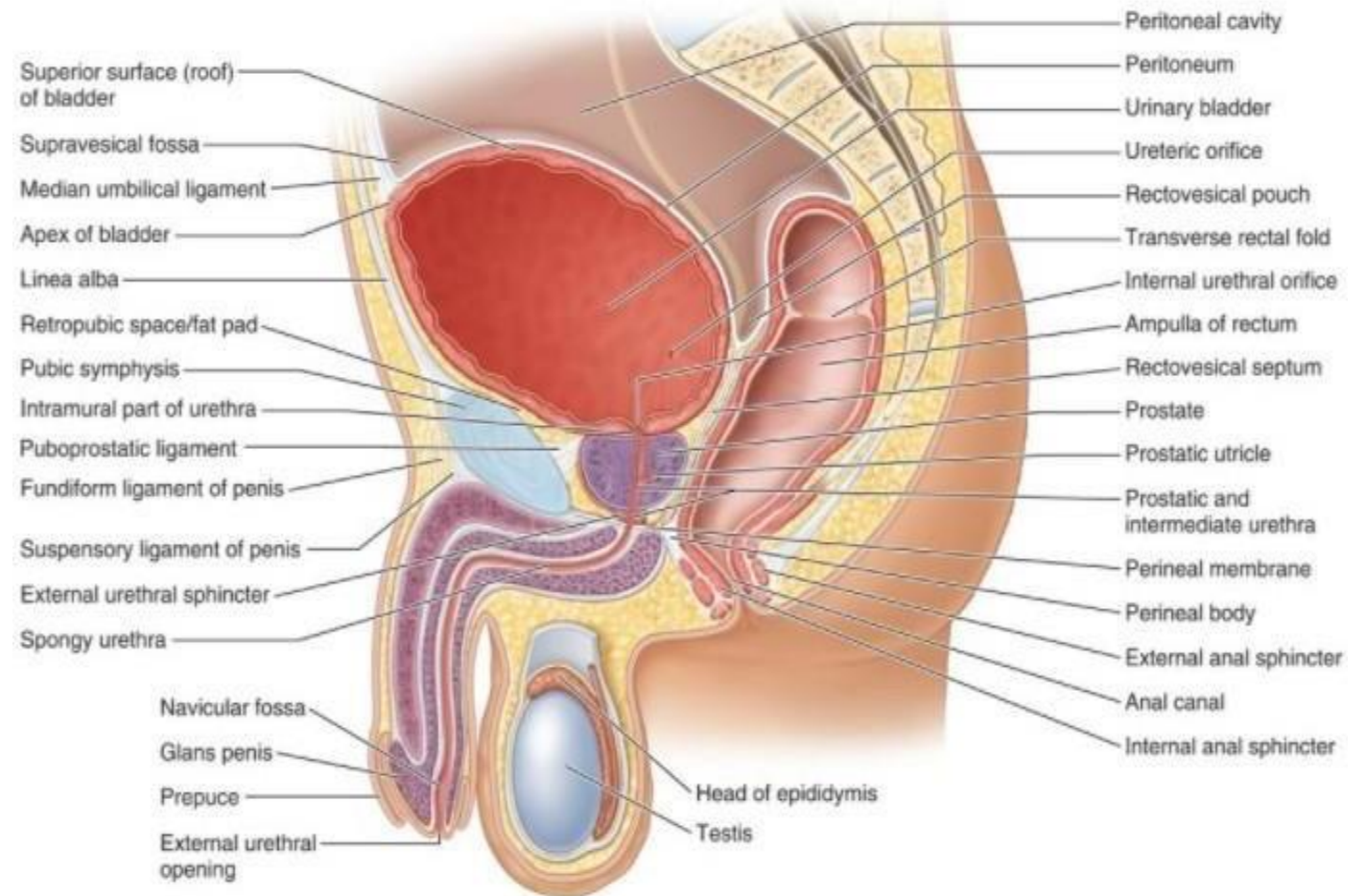
PELVIC GASTROINTESTINAL ORGANS

- The pelvic gastrointestinal organs include:
 - The terminal part of the sigmoid colon
 - The rectum
 - The anal canal

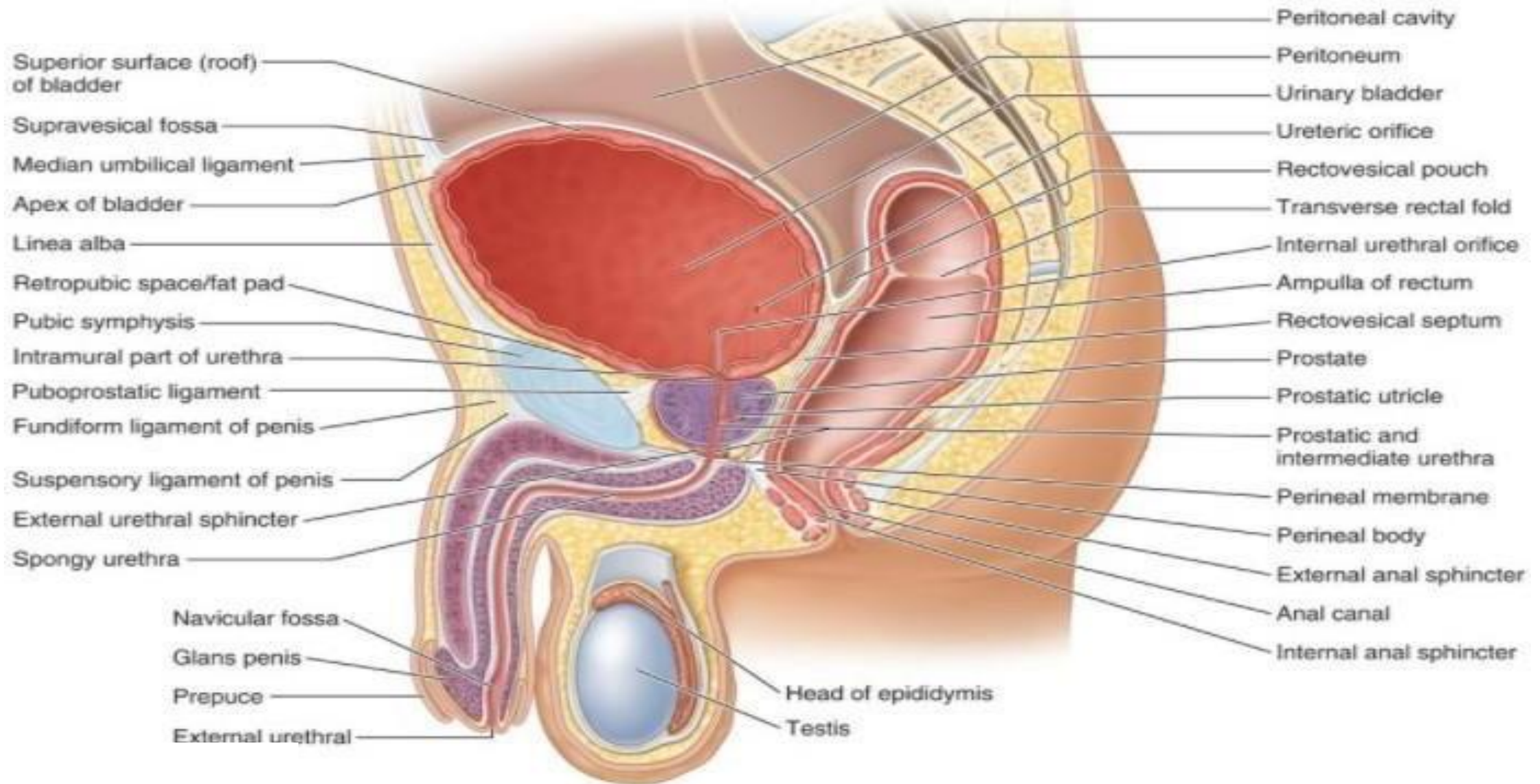


THE RECTUM

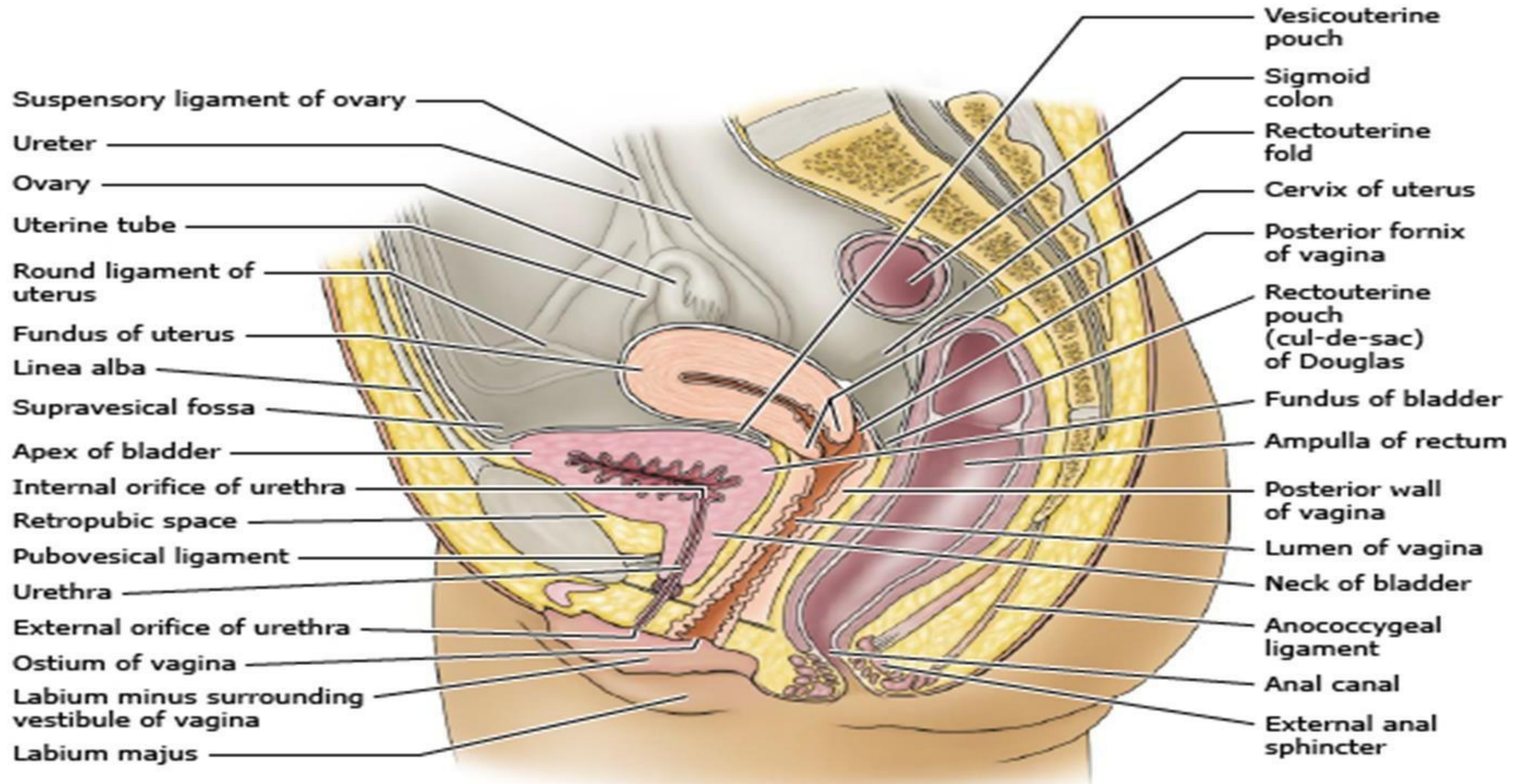
- The most **posterior** organ of the pelvic viscera.
 - Lies anterior to and follows the concave contour of the sacrum.
 - Lies posterior to the fundus of the urinary bladder, the terminal parts of the ureters, ductus deferens, seminal glands and prostate.
- The rectovesical pouch lies between the fundus of the bladder and the ampulla of the rectum in men.
- When a man is upright or supine the rectovesical pouch is the lowest part of his peritoneal cavity so fluids accumulate in this pouch.



LET US RECAP!



LET US RECAP!



Thank you!!!!